

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

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FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
AC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] access control policy that: a. Addresses purpose, scope, and objectives; b. Defines the types of accounts allowed and specifically prohibited for use within the system; c. Assign account managers; c. Require [Assignment: organization-defined prerequisites and criteria] for group and role.	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
AC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] access control policy that: a. Addresses purpose, scope, and objectives; b. Defines the types of accounts allowed and specifically prohibited for use within the system; c. Assign account managers; c. Require [Assignment: organization-defined prerequisites and criteria] for group and role.	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCR), including policies, standards and procedures, at	5	
AC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] access control policy that: a. Addresses purpose, scope, and objectives; b. Defines the types of accounts allowed and specifically prohibited for use within the system; c. Assign account managers; c. Require [Assignment: organization-defined prerequisites and criteria] for group and role.	Functional	Subset Of	Identity & Access Management (IAM)	IAC-01	Mechanisms exist to facilitate the implementation of identification and access management controls.	10	
AC-02	Account Management	a. Define and document the types of accounts allowed and specifically prohibited for use within the system; b. Assign account managers; c. Require [Assignment: organization-defined prerequisites and criteria] for group and role.	Functional	Intersects With	Termination of Employment	IAC-07.2	Mechanisms exist to revoke user access rights in a timely manner, upon termination of employment or contract.	5	
AC-02	Account Management	a. Define and document the types of accounts allowed and specifically prohibited for use within the system; b. Assign account managers; c. Require [Assignment: organization-defined prerequisites and criteria] for group and role.	Functional	Intersects With	Account Management	IAC-15	Mechanisms exist to: (1) Define authorized system account types; (2) Define prohibited system account	5	
AC-02	Account Management	a. Define and document the types of accounts allowed and specifically prohibited for use within the system; b. Assign account managers; c. Require [Assignment: organization-defined prerequisites and criteria] for group and role.	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5	
AC-02	Account Management	a. Define and document the types of accounts allowed and specifically prohibited for use within the system; b. Assign account managers; c. Require [Assignment: organization-defined prerequisites and criteria] for group and role.	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5	
AC-02(01)	Account Management Automated System Account Management	Support the management of system accounts using [Assignment: organization-defined automated mechanisms].	Functional	Intersects With	Automated System Account Management (Directory Services)	IAC-15.1	Automated mechanisms exist to support the management of system accounts (e.g., directory services).	5	
AC-02(02)	Account Management Automated Temporary and Emergency Account	Automatically [Selection (one): remove; disable] temporary and emergency accounts after [Assignment: organization-defined time period for each type of account].	Functional	Equal	Removal of Temporary / Emergency Accounts	IAC-15.2	Automated mechanisms exist to disable or remove temporary and emergency accounts after an organization-defined time period for	10	
AC-02(03)	Account Management Disable Accounts	Disable accounts within [Assignment: organization-defined time period] when the accounts: a. Have expired; b. Are no longer associated with a user or individual; c. Are in violation of organizational policy; or d. Have been inactive for [Assignment: organization-defined time period].	Functional	Equal	Disable Inactive Accounts	IAC-15.3	Automated mechanisms exist to disable inactive accounts after an organization-defined time period.	10	
AC-02(04)	Account Management Automated Audit Actions	Automatically audit account creation, modification, enabling, disabling, and removal actions.	Functional	Equal	Automated Audit Actions	IAC-15.4	Automated mechanisms exist to audit account creation, modification, enabling, disabling and removal actions and notify organization.	10	
AC-02(05)	Account Management Inactivity Logout	Require that users log out when [Assignment: organization-defined time period of expected inactivity or description of when to log out].	Functional	Equal	Session Lock	IAC-24	Mechanisms exist to initiate a session lock after an organization-defined time period of inactivity, or upon receiving a request from a user and	10	
AC-02(13)	Account Management Disable Accounts for High-risk Individuals	Disable accounts of individuals within [Assignment: organization-defined time period] of discovery of [Assignment: organization-defined significant risks].	Functional	Intersects With	High-Risk Terminations	HRS-09.2	Mechanisms exist to expedite the process of removing "high risk" individual's access to Technology Assets, Applications, Services and/or	5	
AC-02(13)	Account Management Disable Accounts for High-risk Individuals	Disable accounts of individuals within [Assignment: organization-defined time period] of discovery of [Assignment: organization-defined significant risks].	Functional	Intersects With	Account Disabling for High Risk Individuals	IAC-15.6	Mechanisms exist to disable accounts immediately upon notification for users posing a significant risk to the organization.	5	
AC-03	Access Enforcement	Enforce approved authorizations for logical access to information and system resources in accordance with applicable access control policies.	Functional	Intersects With	Access Enforcement	IAC-20	Mechanisms exist to enforce Logical Access Control (LAC) permissions that conform to the principle of "least privilege."	5	
AC-03	Access Enforcement	Enforce approved authorizations for logical access to information and system resources in accordance with applicable access control policies.	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5	
AC-03	Access Enforcement	Enforce approved authorizations for logical access to information and system resources in accordance with applicable access control policies.	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5	
AC-04	Information Flow Enforcement	Enforce approved authorizations for controlling the flow of information within the system and between connected systems based on [Assignment: organization-defined information flow control policies].	Functional	Equal	Data Flow Enforcement - Access Control Lists (ACLs)	NET-04	Mechanisms exist to implement and govern Access Control Lists (ACLs) to provide data flow enforcement that explicitly restrict network traffic to	10	
AC-05	Separation of Duties	a. Identify and document [Assignment: organization-defined duties of individuals requiring separation]; and b. Define system access authorizations to support separation of duties.	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5	
AC-05	Separation of Duties	a. Identify and document [Assignment: organization-defined duties of individuals requiring separation]; and b. Define system access authorizations to support separation of duties.	Functional	Intersects With	Dual Authorization for Change	CHG-04.3	Mechanisms exist to enforce a two-person rule for implementing changes to critical Technology Assets, Applications, Services and/or	5	
AC-05	Separation of Duties	a. Identify and document [Assignment: organization-defined duties of individuals requiring separation]; and b. Define system access authorizations to support separation of duties.	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5	
AC-05	Separation of Duties	a. Identify and document [Assignment: organization-defined duties of individuals requiring separation]; and b. Define system access authorizations to support separation of duties.	Functional	Intersects With	Separation of Duties (SoD)	HRS-11	Mechanisms exist to implement and maintain Separation of Duties (SoD) to prevent potential inappropriate activity without collusion.	5	
AC-06	Least Privilege	Employ the principle of least privilege, allowing only authorized accesses for users (or processes acting on behalf of users) that are necessary to accomplish assigned organizational tasks.	Functional	Intersects With	Least Privilege	IAC-21	Mechanisms exist to utilize the concept of least privilege, allowing only authorized access to processes necessary to accomplish assigned	5	
AC-06	Least Privilege	Employ the principle of least privilege, allowing only authorized accesses for users (or processes acting on behalf of users) that are necessary to accomplish assigned organizational tasks.	Functional	Intersects With	Access Enforcement	IAC-20	Mechanisms exist to enforce Logical Access Control (LAC) permissions that conform to the principle of "least privilege."	5	
AC-06(01)	Least Privilege Authorize Access to Security Functions	Authorize access for [Assignment: organization-defined individuals or roles] to: a. [Assignment: organization-defined security functions (deployed in hardware, software, and firmware)]; and b. [Assignment: organization-defined security-	Functional	Equal	Authorize Access to Security Functions	IAC-21.1	Mechanisms exist to limit access to security functions to explicitly-authorized privileged users.	10	

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AC-06(02)	Least Privilege Non-privileged Access for Nonsecurity Functions	Require that users of system accounts (or roles) with access to [Assignment: organization-defined security functions or security-relevant information] use non-privileged accounts or roles, when accessing nonsecurity functions.	Functional	Equal	Non-Privileged Access for Non-Security Functions	IAC-21.2	Mechanisms exist to prohibit privileged users from using privileged accounts, while performing non-security functions.	10	
AC-06(05)	Least Privilege Privileged Accounts	Restrict privileged accounts on the system to [Assignment: organization-defined personnel or roles].	Functional	Equal	Management Approval For Privileged Accounts	IAC-21.3	Mechanisms exist to restrict the assignment of privileged accounts to management-approved personnel and/or roles.	10	
AC-06(07)	Least Privilege Review of User Privileges	a. Review [Assignment: organization-defined frequency] the privileges assigned to [Assignment: organization-defined roles or classes of users] to validate the need for such privileges; andb. Reassign or remove privileges, if necessary, to correctly	Functional	Equal	Periodic Review of Account Privileges	IAC-17	Mechanisms exist to periodically review the privileges assigned to individuals and service accounts to validate the need for such privileges.	10	
AC-06(09)	Least Privilege Log Use of Privileged Functions	Log the execution of privileged functions.	Functional	Equal	Auditing Use of Privileged Functions	IAC-21.4	Mechanisms exist to audit the execution of privileged functions.	10	
AC-06(10)	Least Privilege Prohibit Non-privileged Users from Executing Privileged Functions	Prevent non-privileged users from executing privileged functions.	Functional	Equal	Prohibit Non-Privileged Users from Executing Privileged Functions	IAC-21.5	Mechanisms exist to prevent non-privileged users from executing privileged functions to include disabling, circumventing or altering	10	
AC-07	Unsuccessful Logon Attempts	a. Enforce a limit of [Assignment: organization-defined number] consecutive invalid logon attempts by a user during a [Assignment: organization-defined time period]; andb. Automatically [Selection (one or more): lock the account or	Functional	Equal	Account Lockout	IAC-22	Mechanisms exist to enforce a limit for consecutive invalid logon attempts by a user during an organization-defined time period and automatically	10	
AC-08	System Use Notification	a. Display [Assignment: organization-defined system use notification message or banner] to users before granting access to the system that provides privacy and security notices consistent with applicable laws, executive orders, directives	Functional	Equal	System Use Notification (Logon Banner)	SEA-18	Mechanisms exist to utilize system use notification / logon banners that display an approved system use notification message or banner	10	
AC-11	Device Lock	a. Prevent further access to the system by [Selection (one or more): initiating a device lock after [Assignment: organization-defined time period] of inactivity; requiring the user to initiate a device lock before leaving the system unattended]; andb.	Functional	Intersects With	Session Lock	IAC-24	Mechanisms exist to initiate a session lock after an organization-defined time period of inactivity, or upon receiving a request from a user and	5	
AC-11(01)	Device Lock Pattern-hiding Displays	Conceal, via the device lock, information previously visible on the display with a publicly viewable image.	Functional	Equal	Pattern-Hiding Displays	IAC-24.1	Mechanisms exist to implement pattern-hiding displays to conceal information previously visible on the display during the session lock.	10	
AC-12	Session Termination	Automatically terminate a user session after [Assignment: organization-defined conditions or trigger events requiring session disconnect].	Functional	Equal	Session Termination	IAC-25	Automated mechanisms exist to log out users, both locally on the network and for remote sessions, at the end of the session or after an organization-	10	
AC-14	Permitted Actions Without Identification or Authentication	a. Identify [Assignment: organization-defined user actions] that can be performed on the system without identification or authentication consistent with organizational mission and business functions; andb. Document and provide supporting	Functional	Equal	Permitted Actions Without Identification or Authorization	IAC-26	Mechanisms exist to identify and document the supporting rationale for specific user actions that can be performed on a system without	10	
AC-17	Remote Access	a. Establish and document usage restrictions, configuration/connection requirements, and implementation guidance for each type of remote access allowed; andb. Authorize each type of remote access to the system prior to	Functional	Intersects With	Remote Access	NET-14	Mechanisms exist to define, control and review organization-approved, secure remote access methods.	5	
AC-17(01)	Remote Access Monitoring and Control	Employ automated mechanisms to monitor and control remote access methods.	Functional	Equal	Automated Monitoring & Control	NET-14.1	Automated mechanisms exist to monitor and control remote access sessions.	10	
AC-17(02)	Remote Access Protection of Confidentiality and Integrity Using	Implement cryptographic mechanisms to protect the confidentiality and integrity of remote access sessions.	Functional	Equal	Protection of Confidentiality / Integrity Using Encryption	NET-14.2	Cryptographic mechanisms exist to protect the confidentiality and integrity of remote access sessions (e.g., VPN).	10	
AC-17(03)	Remote Access Managed Access Control Points	Route remote accesses through authorized and managed network access control points.	Functional	Equal	Managed Access Control Points	NET-14.3	Mechanisms exist to route all remote accesses through managed network access control points (e.g., VPN concentrator).	10	
AC-17(04)	Remote Access Privileged Commands and Access	a. Authorize the execution of privileged commands and access to security-relevant information via remote access only in a format that provides assessable evidence and for the following needs: [Assignment: organization-defined needs]; andb.	Functional	Equal	Remote Privileged Commands & Sensitive Data Access	NET-14.4	Mechanisms exist to restrict the execution of privileged commands and access to security-relevant information via remote access only.	10	
AC-17(09)	Remote Access Disconnect or Disable Access	Provide the capability to disconnect or disable remote access to the system within [Assignment: organization-defined time period].	Functional	Equal	Expeditious Disconnect / Disable Capability	NET-14.8	Mechanisms exist to provide the capability to expeditiously disconnect or disable a user's remote access session.	10	
AC-17(10)	Remote Access Authenticate Remote Commands	Implement [Assignment: organization-defined mechanisms] to authenticate [Assignment: organization-defined remote commands].	Functional	No Relationship	N/A	N/A	N/A	0	
AC-18	Wireless Access	a. Establish configuration requirements, connection requirements, and implementation guidance for each type of wireless access; andb. Authorize each type of wireless access to the system prior to allowing such connections.	Functional	Intersects With	Wireless Networking	NET-15	Mechanisms exist to control authorized wireless usage and monitor for unauthorized wireless access.	5	
AC-18	Wireless Access	a. Establish configuration requirements, connection requirements, and implementation guidance for each type of wireless access; andb. Authorize each type of wireless access to the system prior to allowing such connections.	Functional	Intersects With	Wireless Access Authentication & Encryption	CRY-07	Mechanisms exist to protect the confidentiality and integrity of wireless networking technologies by implementing authentication and	5	
AC-18(01)	Wireless Access Authentication and Encryption	Protect wireless access to the system using authentication of [Selection (one or more): users; devices] and encryption.	Functional	Equal	Authentication & Encryption	NET-15.1	Mechanisms exist to secure Wi-Fi (e.g., IEEE 802.11) and prevent unauthorized access by: (1) Authenticating devices to the	10	
AC-18(03)	Wireless Access Disable Wireless Networking	Disable, when not intended for use, wireless networking capabilities embedded within system components prior to issuance and deployment.	Functional	Equal	Disable Wireless Networking	NET-15.2	Mechanisms exist to disable unnecessary wireless networking capabilities that are internally embedded within system	10	
AC-19	Access Control for Mobile Devices	a. Establish configuration requirements, connection requirements, and implementation guidance for organization-controlled mobile devices, to include when such devices are outside of controlled areas; andb. Authorize the connection of	Functional	Equal	Access Control For Mobile Devices	MDM-02	Mechanisms exist to enforce access control requirements for the connection of mobile devices to organizational Technology Assets.	10	
AC-19(05)	Access Control for Mobile Devices Full Device or Container-based Encryption	Employ [Selection: full-device encryption; container-based encryption] to protect the confidentiality and integrity of information on [Assignment: organization-defined mobile devices].	Functional	Equal	Full Device & Container-Based Encryption	MDM-03	Cryptographic mechanisms exist to protect the confidentiality and integrity of information on mobile devices through full-device or	10	
AC-20	Use of External Systems	a. [Selection (one or more): Establish [Assignment: organization-defined terms and conditions]; identify [Assignment: organization-defined controls asserted to be implemented on external systems]] consistent with the trust	Functional	Equal	Use of External Technology Assets, Applications and/or Services (TAAS)	DCH-13	Mechanisms exist to govern how external parties, including Technology Assets, Applications and/or Services (TAAS), are used to accomplish	10	

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
AC-20(01)	Use of External Systems Limits on Authorized Use	Permit authorized individuals to use an external system to access the system or to process, store, or transmit organization-controlled information only after: a. Verification of the implementation of controls on the external system as	Functional	Equal	Limits of Authorized Use	DCH-13.1	Mechanisms exist to prohibit external parties, including Technology Assets, Applications and/or Services (TAAS), from storing, processing and	10	
AC-20(02)	Use of External Systems Portable Storage Devices — Restricted Use	Restrict the use of organization-controlled portable storage devices by authorized individuals on external systems using [Assignment: organization-defined restrictions].	Functional	Intersects With	Portable Storage Devices	DCH-13.2	Mechanisms exist to restrict or prohibit the use of portable storage devices by users on external systems.	5	
AC-21	Information Sharing	a. Enable authorized users to determine whether access authorizations assigned to a sharing partner match the information's access and use restrictions for [Assignment: organization-defined information sharing circumstances where]; a. Designate individuals authorized to make information	Functional	Intersects With	Information Sharing With Third Parties	PRI-07	Mechanisms exist to disclose Personal Data (PD) to third-parties only for the purposes identified in the data privacy notice and with the	5	
AC-22	Publicly Accessible Content	publicly accessible;b. Train authorized individuals to ensure that publicly accessible information does not contain nonpublic information;c. Review the proposed content of	Functional	Equal	Publicly Accessible Content	DCH-15	Mechanisms exist to control publicly-accessible content.	10	
AT-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level]; awareness and training policy that a. Addresses	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
AT-02	Literacy Training and Awareness	a. Provide security and privacy literacy training to system users (including managers, senior executives, and contractors):1. As part of initial training for new users and [Assignment: organization-defined frequency]; thereafter: and 2. When	Functional	Equal	Security, Compliance & Resilience Awareness Training	SAT-02	Mechanisms exist to provide all employees and contractors appropriate security, compliance and resilience awareness education and	10	
AT-02(02)	Literacy Training and Awareness Insider Threat	Provide literacy training on recognizing and reporting potential indicators of insider threat.	Functional	Equal	Insider Threat Awareness	THR-05	Mechanisms exist to utilize security awareness training on recognizing and reporting potential indicators of insider threat	10	
AT-02(03)	Literacy Training and Awareness Social Engineering and Mining	Provide literacy training on recognizing and reporting potential and actual instances of social engineering and social mining.	Functional	Equal	Social Engineering & Mining	SAT-02.2	Mechanisms exist to include awareness training on recognizing and reporting potential and actual instances of social engineering and	10	
AT-02(04)	Literacy Training and Awareness Suspicious Communications and	Provide literacy training on recognizing suspicious communications and anomalous behavior in organizational systems using [Assignment: organization-defined indicators of malicious code];	Functional	Intersects With	Suspicious Communications & Anomalous System Behavior	SAT-03.2	Mechanisms exist to provide training to personnel on organization-defined indicators of malware to recognize suspicious communications and	5	
AT-03	Role-based Training	a. Provide role-based security and privacy training to personnel with the following roles and responsibilities: [Assignment: organization-defined roles and responsibilities];1. Before authorizing access to the system, information, or performing	Functional	Intersects With	Role-Based Security, Compliance & Resilience Training	SAT-03	Mechanisms exist to provide role-based security, compliance and resilience-related training: (1) Before authorizing access to the	5	
AT-04	Training Records	a. Document and monitor information security and privacy training activities, including security and privacy awareness training and specific role-based security and privacy training; and b. Retain individual training records for [Assignment:	Functional	Equal	Security, Compliance & Resilience Training Records	SAT-04	Mechanisms exist to document, retain and monitor individual training activities, including: (1) Initial security, compliance and	10	
AU-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level]; audit and accountability policy that a. Addresses	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRP), including policies, standards and procedures	5	
AU-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level]; audit and accountability policy that a. Addresses	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
AU-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level]; audit and accountability policy that a. Addresses	Functional	Subset Of	Continuous Monitoring	MON-01	Mechanisms exist to facilitate the implementation of enterprise-wide monitoring controls.	10	
AU-02	Event Logging	a. Identify the types of events that the system is capable of logging in support of the audit function; [Assignment: organization-defined event types that the system is capable of logging];b. Coordinate the event logging function with other	Functional	Intersects With	Security Event Monitoring	MON-01.8	Mechanisms exist to review event logs on an ongoing basis and escalate incidents in accordance with	5	
AU-02	Event Logging	a. Identify the types of events that the system is capable of logging in support of the audit function; [Assignment: organization-defined event types that the system is capable of logging];b. Coordinate the event logging function with other	Functional	Intersects With	Centralized Collection of Security Event Logs	MON-02	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support the centralized collection of security	5	
AU-03	Content of Audit Records	Ensure that audit records contain information that establishes the following:a. What type of event occurred;b. When the event occurred;c. Where the event occurred;d. Source of the event;e. Outcome of the event; and f. Identity of any individuals.	Functional	Equal	Content of Event Logs	MON-03	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to produce event logs that contain sufficient	10	
AU-03(01)	Content of Audit Records Additional Audit Information	Generate audit records containing the following additional information: [Assignment: organization-defined additional information].	Functional	Intersects With	Sensitive Event Log Information	MON-03.1	Mechanisms exist to protect sensitive and/or regulated data contained in log files.	5	
AU-04	Audit Log Storage Capacity	Allocate audit log storage capacity to accommodate [Assignment: organization-defined audit log retention requirements].	Functional	Equal	Event Log Storage Capacity	MON-04	Mechanisms exist to allocate and proactively manage sufficient event log storage capacity to reduce the likelihood of such capacity being	10	
AU-04(01)	Audit Log Storage Capacity Transfer to Alternate Storage	Transfer audit logs [Assignment: organization-defined frequency] to a different system, system component, or media other than the system or system component conducting the logging	Functional	Intersects With	Event Log Backup on Separate Physical Systems / Components	MON-08.1	Mechanisms exist to back up event logs onto a physically different system or system component than the Security Incident Event Manager	5	
AU-05	Response to Audit Logging Process Failures	a. Alert [Assignment: organization-defined personnel or roles] within [Assignment: organization-defined time period] in the event of an audit logging process failure; and b. Take the following additional actions: [Assignment: organization-defined	Functional	Equal	Response To Event Log Processing Failures	MON-05	Mechanisms exist to alert appropriate personnel in the event of a log processing failure and take actions to remedy the disruption	10	
AU-06	Audit Record Review, Analysis, and Reporting	a. Review and analyze system audit records [Assignment: organization-defined frequency] for indications of [Assignment: organization-defined inappropriate or unusual activity] and the potential impact of the inappropriate or unusual activity ;b.	Functional	Intersects With	Centralized Collection of Security Event Logs	MON-02	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support the centralized collection of security	5	
AU-06	Audit Record Review, Analysis, and Reporting	a. Review and analyze system audit records [Assignment: organization-defined frequency] for indications of [Assignment: organization-defined inappropriate or unusual activity] and the potential impact of the inappropriate or unusual activity ;b.	Functional	Intersects With	Audit Level Adjustments	MON-02.6	Mechanisms exist to adjust the level of audit review, analysis and reporting based on evolving threat information from law enforcement industry	5	
AU-06(01)	Audit Record Review, Analysis, and Reporting Automated Process Integration	Integrate audit record review, analysis, and reporting processes using [Assignment: organization-defined automated mechanisms].	Functional	Intersects With	Sensitive Event Log Information	MON-03.1	Mechanisms exist to protect sensitive and/or regulated data contained in log files.	5	
AU-06(03)	Audit Record Review, Analysis, and Reporting Correlate Audit Record	Analyze and correlate audit records across different repositories to gain organization-wide situational awareness.	Functional	Intersects With	Correlate Monitoring Information	MON-02.1	Automated mechanisms exist to correlate both technical and non-technical information from across the enterprise by a Security Incident	5	

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FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
AU-07	Audit Record Reduction and Report Generation	Provide and implement an audit record reduction and report generation capability that: a. Supports on-demand audit record review, analysis, and reporting requirements and after-the-fact investigations of incidents; and b. Does not alter the original data.	Functional	Intersects With	Monitoring Reporting	MON-06	Mechanisms exist to provide an event log report generation capability to aid in detecting and assessing anomalous activities.	5	
AU-07(01)	Audit Record Reduction and Report Generation Automatic Processing	Provide and implement the capability to process, sort, and search audit records for events of interest based on the following content: [Assignment: organization-defined fields within audit records].	Functional	Intersects With	Monitoring Reporting	MON-06	Mechanisms exist to provide an event log report generation capability to aid in detecting and assessing anomalous activities.	5	
AU-08	Time Stamps	a. Use internal system clocks to generate time stamps for audit records; and b. Record time stamps for audit records that meet [Assignment: organization-defined granularity of time measurement] and that use Coordinated Universal Time, have a. Use internal system clocks to generate time stamps for audit records; and b. Record time stamps for audit records that meet [Assignment: organization-defined granularity of time measurement] and that use Coordinated Universal Time, have	Functional	Intersects With	Clock Synchronization	SEA-20	Mechanisms exist to utilize time-synchronization technology to synchronize all critical system clocks.	5	
AU-08	Time Stamps	a. Use internal system clocks to generate time stamps for audit records; and b. Record time stamps for audit records that meet [Assignment: organization-defined granularity of time measurement] and that use Coordinated Universal Time, have	Functional	Intersects With	Time Stamps	MON-07	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to use an event log report generation capability to aid in detecting and assessing anomalous activities.	5	
AU-09	Protection of Audit Information	a. Protect audit information and audit logging tools from unauthorized access, modification, and deletion; and b. Alert [Assignment: organization-defined personnel or roles] upon detection of unauthorized access, modification, or deletion of audit information.	Functional	Equal	Protection of Event Logs	MON-08	Mechanisms exist to protect event logs and audit tools from unauthorized access, modification and deletion.	10	
AU-09(04)	Protection of Audit Information Access by Subset of Privileged Users	Authorize access to management of audit logging functionality to only [Assignment: organization-defined subset of privileged users or roles].	Functional	Equal	Access by Subset of Privileged Users	MON-08.2	Mechanisms exist to restrict access to the management of event logs to privileged users with a specific business need.	10	
AU-11	Audit Record Retention	Retain audit records for [Assignment: organization-defined time period consistent with records retention policy] to provide support for after-the-fact investigations of incidents and to meet regulatory and organizational information retention requirements.	Functional	Equal	Event Log Retention	MON-10	Mechanisms exist to retain event logs for a time period consistent with records retention requirements to provide support for after-the-fact investigations of incidents.	10	
AU-12	Audit Record Generation	a. Provide audit record generation capability for the event types the system is capable of auditing as defined in AU-2a on [Assignment: organization-defined system components]; b. Allow [Assignment: organization-defined personnel or roles] to a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] assessment, authorization, and monitoring	Functional	Intersects With	Monitoring Reporting	MON-06	Mechanisms exist to provide an event log report generation capability to aid in detecting and assessing anomalous activities.	5	
CA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] assessment, authorization, and monitoring	Functional	Subset Of	Information Assurance (IA) Operations	IAO-01	Mechanisms exist to facilitate the implementation of security, compliance and resilience assessment and authorization.	10	
CA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] assessment, authorization, and monitoring	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCR), including policies, standards and procedures, at least annually.	5	
CA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] assessment, authorization, and monitoring	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient systems.	5	
CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted; b. Develop a control assessment plan that describes the scope of the assessment including: 1. Controls and control enhancements under	Functional	Intersects With	Functional Review Of Security, Compliance & Resilience Controls	CPL-03.2	Mechanisms exist to regularly review Technology Assets, Applications and/or Services (TAAS) for adherence to the organization's security, compliance and resilience requirements.	5	
CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted; b. Develop a control assessment plan that describes the scope of the assessment including: 1. Controls and control enhancements under	Functional	Intersects With	Technical Verification	IAO-06	Mechanisms exist to perform Information Assurance Program (IAP) activities to evaluate the design, implementation and effectiveness of	5	
CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted; b. Develop a control assessment plan that describes the scope of the assessment including: 1. Controls and control enhancements under	Functional	Intersects With	Security, Compliance & Resilience In Project Management	PRM-04	Mechanisms exist to assess security, compliance and resilience controls as part of Technology Assets, Applications and/or Services (TAAS).	5	
CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted; b. Develop a control assessment plan that describes the scope of the assessment including: 1. Controls and control enhancements under	Functional	Intersects With	Assessments	IAO-02	Mechanisms exist to formally assess the security, compliance and resilience controls in Technology Assets, Applications and/or Services (TAAS).	5	
CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted; b. Develop a control assessment plan that describes the scope of the assessment including: 1. Controls and control enhancements under	Functional	Intersects With	Control Conformity Monitoring	CPL-03	Mechanisms exist to validate that Technology Assets, Applications, Services and/or Data (TAASD) conform to the organization's security, compliance and resilience requirements.	5	
CA-02(01)	Control Assessments Independent Assessors	Employ independent assessors or assessment teams to conduct control assessments.	Functional	Equal	Assessor Independence	IAO-02.1	Mechanisms exist to ensure assessors or assessment teams have the appropriate independence to conduct security, compliance and/or resilience assessments.	10	
CA-03	Information Exchange	a. Approve and manage the exchange of information between the system and other systems using [Selection (one or more): interconnection security agreements; information exchange security agreements; memoranda of understanding or	Functional	Intersects With	Interconnection Security Agreements (ISAs)	NET-05	Mechanisms exist to authorize connections from systems to other systems using Interconnection Security Agreements (ISAs), or similar	5	
CA-05	Plan of Action and Milestones	a. Develop a plan of action and milestones for the system to document the planned remediation actions of the organization to correct weaknesses or deficiencies noted during the assessment of the controls and to reduce or eliminate known	Functional	Intersects With	Capabilities Deficiency Tracking	IAO-05	Mechanisms exist to govern identified deficiencies (e.g., Plan of Action and Milestones (POA&M) or similar methodology) that formally	5	
CA-06	Authorization	a. Assign a senior official as the authorizing official for the system; b. Assign a senior official as the authorizing official for common controls available for inheritance by organizational systems; c. Ensure that the authorizing official for the system	Functional	Equal	Security Authorization	IAO-07	Mechanisms exist to ensure Technology Assets, Applications and/or Services (TAAS) are officially	10	
CA-07	Continuous Monitoring	Develop a system-level continuous monitoring strategy and implement continuous monitoring in accordance with the organization-level continuous monitoring strategy that includes: a. Establishing the following system-level metrics to	Functional	Intersects With	Security, Compliance & Resilience Controls Oversight	CPL-02	Mechanisms exist to provide a security, compliance and resilience controls oversight function that reports to the organization's executive	5	
CA-07(01)	Continuous Monitoring Independent Assessment	Employ independent assessors or assessment teams to monitor the controls in the system on an ongoing basis.	Functional	Intersects With	Independent Assessors	CPL-03.1	Mechanisms exist to utilize independent assessors to evaluate security, compliance and resilience controls.	5	
CA-07(04)	Continuous Monitoring Risk Monitoring	Ensure risk monitoring is an integral part of the continuous monitoring strategy that includes the following: a. Effectiveness monitoring; b. Compliance monitoring; and c. Change monitoring.	Functional	Equal	Risk Monitoring	RSK-11	Mechanisms exist to ensure risk monitoring as an integral part of the continuous monitoring strategy that includes monitoring the effectiveness	10	
CA-09	Internal System Connections	a. Authorize internal connections of [Assignment: organization-defined system components or classes of components] to the system; b. Document, for each internal connection, the interface characteristics, security and privacy requirements	Functional	Equal	Internal System Connections	NET-05.2	Mechanisms exist to control internal system connections through authorizing internal connections of systems and documenting for each	10	
CM-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] configuration management policy that: a	Functional	Subset Of	Configuration Management Program	CFG-01	Mechanisms exist to facilitate the implementation of configuration management controls.	10	

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
CM-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] configuration management policy that a.	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures necessary for secure, compliant and resilient	5	
CM-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] configuration management policy that a.	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
CM-02	Baseline Configuration	a. Develop, document, and maintain under configuration control, a current baseline configuration of the system; andb. Review and update the baseline configuration of the system:1. [Assignment: organization-defined frequency];2. When	Functional	Intersects With	Reviews & Updates	CFG-02.1	Mechanisms exist to review and update baseline configurations: (1) At least annually; (2) When required due to an	5	
CM-02	Baseline Configuration	a. Develop, document, and maintain under configuration control, a current baseline configuration of the system; andb. Review and update the baseline configuration of the system:1. [Assignment: organization-defined frequency];2. When	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for	5	
CM-02(02)	Baseline Configuration Automation Support for Accuracy and Baseline	Maintain the currency, completeness, accuracy, and availability of the baseline configuration of the system using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Automated Central Management & Verification	CFG-02.2	Automated mechanisms exist to govern and report on baseline configurations of Technology Assets, Applications, Services and/or Data (TAASD)	10	
CM-02(03)	Baseline Configuration Retention of Previous Configurations	Retain [Assignment: organization-defined number] of previous versions of baseline configurations of the system to support rollback.	Functional	Equal	Retention Of Previous Configurations	CFG-02.3	Mechanisms exist to retain previous versions of baseline configuration to support roll back.	10	
CM-02(07)	Baseline Configuration Configure Systems and Components for	a. Issue [Assignment: organization-defined systems or system components] with [Assignment: organization-defined configurations] to individuals traveling to locations that the organization deems to be of significant risk; andb. Apply the	Functional	Equal	Configure Technology Assets, Applications and/or Services (TAAS) for High-Risk Areas	CFG-02.5	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) utilized in high-risk areas with appropriate	10	
CM-03	Configuration Change Control	a. Determine and document the types of changes to the system that are configuration-controlled;b. Review proposed configuration-controlled changes to the system and approve or disapprove such changes with explicit consideration for	Functional	Subset Of	Change Management Program	CHG-01	Mechanisms exist to facilitate the implementation of a change management program.	10	
CM-03	Configuration Change Control	a. Determine and document the types of changes to the system that are configuration-controlled;b. Review proposed configuration-controlled changes to the system and approve or disapprove such changes with explicit consideration for	Functional	Intersects With	Configuration Change Control	CHG-02	Mechanisms exist to govern the technical configuration change control processes.	5	
CM-03(02)	Configuration Change Control Testing, Validation, and Documentation of	Test, validate, and document changes to the system before finalizing the implementation of the changes.	Functional	Intersects With	Control Functionality Verification	CHG-06	Mechanisms exist to verify the functionality of security, compliance and resilience controls following implemented changes to ensure	5	
CM-03(04)	Configuration Change Control Security and Privacy Representatives	Require [Assignment: organization-defined security and privacy representatives] to be members of the [Assignment: organization-defined configuration change control element].	Functional	Equal	Security, Compliance & Resilience Representative for Asset Lifecycle	CHG-02.3	Mechanisms exist to include a cybersecurity and/or data protection representative in the configuration	10	
CM-04	Impact Analyses	Analyze changes to the system to determine potential security and privacy impacts prior to change implementation.	Functional	Equal	Security Impact Analysis for Changes	CHG-03	Mechanisms exist to analyze proposed changes for potential security impacts, prior to the implementation of the change	10	
CM-04(02)	Impact Analyses Verification of Controls	After system changes, verify that the impacted controls are implemented correctly, operating as intended, and producing the desired outcome with regard to meeting the security and privacy requirements for the system.	Functional	Equal	Technical Verification	IAO-06	Mechanisms exist to perform Information Assurance Program (IAP) activities to evaluate the design, implementation and effectiveness of	10	
CM-05	Access Restrictions for Change	Define, document, approve, and enforce physical and logical access restrictions associated with changes to the system.	Functional	Intersects With	Governing Access Restriction for Change	END-03.2	Mechanisms exist to define, document, approve and enforce access restrictions associated with	5	
CM-05	Access Restrictions for Change	Define, document, approve, and enforce physical and logical access restrictions associated with changes to the system.	Functional	Intersects With	Access Restriction For Change	CHG-04	Mechanisms exist to enforce configuration restrictions in an effort to restrict the ability of users to conduct unauthorized changes	5	
CM-06	Configuration Settings	a. Establish and document configuration settings for components employed within the system that reflect the most restrictive mode consistent with operational requirements using [Assignment: organization-defined common secure	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for	5	
CM-06	Configuration Settings	a. Establish and document configuration settings for components employed within the system that reflect the most restrictive mode consistent with operational requirements using [Assignment: organization-defined common secure	Functional	Intersects With	Approved Configuration Deviations	CFG-02.7	Mechanisms exist to document, assess risk and approve or deny deviations to standardized configurations	5	
CM-07	Least Functionality	a. Configure the system to provide only [Assignment: organization-defined mission essential capabilities]; andb. Prohibit or restrict the use of the following functions, ports, protocols, software, and/or services: [Assignment:	Functional	Equal	Least Functionality	CFG-03	Mechanisms exist to configure systems to provide only essential capabilities by specifically prohibiting or restricting the use of ports	10	
CM-07(01)	Least Functionality Periodic Review	a. Review the system [Assignment: organization-defined frequency] to identify unnecessary and/or nonsecure functions, ports, protocols, software, and services; andb. Disable or remove [Assignment: organization-defined	Functional	Equal	Periodic Review	CFG-03.1	Mechanisms exist to periodically review system configurations to identify and disable unnecessary and/or non-secure functions, ports	10	
CM-07(02)	Least Functionality Prevent Program Execution	Prevent program execution in accordance with [Selection (one or more): [Assignment: organization-defined policies, rules of behavior, and/or access agreements regarding software	Functional	Intersects With	Prevent Program Execution	SEA-06	Automated mechanisms exist to prevent the execution of unauthorized software programs.	5	
CM-07(05)	Least Functionality Authorized Software — Allow-by-exception	a. Identify [Assignment: organization-defined software programs authorized to execute on the system];b. Employ a deny-all, permit-by-exception policy to allow the execution of authorized software programs on the system; andc. Review	Functional	Equal	Explicitly Allow / Deny Applications	CFG-03.3	Mechanisms exist to explicitly allow (allowlist / whitelist) and/or block (denylist / blacklist) applications that are authorized to execute on systems	10	
CM-08	System Component Inventory	a. Develop and document an inventory of system components that:1. Accurately reflects the system;2. Includes all components within the system;3. Does not include duplicate accounting of components or components assigned to any	Functional	Intersects With	Asset Inventories	AST-02	Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data (TAASD)	5	
CM-08	System Component Inventory	a. Develop and document an inventory of system components that:1. Accurately reflects the system;2. Includes all components within the system;3. Does not include duplicate accounting of components or components assigned to any	Functional	Intersects With	Component Duplication Avoidance	AST-02.3	Mechanisms exist to establish and maintain an authoritative source and repository to provide a trusted source and accountability for approved and	5	
CM-08(01)	System Component Inventory Updates During Installation and Removal	Update the inventory of system components as part of component installations, removals, and system updates.	Functional	Equal	Updates During Installations / Removals	AST-02.1	Mechanisms exist to update asset inventories as part of component installations, removals and asset upgrades	10	
CM-08(03)	System Component Inventory Automated Unauthorized Component Detection	a. Detect the presence of unauthorized hardware, software, and firmware components within the system using [Assignment: organization-defined automated mechanisms] [Assignment: organization-defined frequency]; andb. Take the	Functional	Intersects With	Automated Unauthorized Component Detection	AST-02.2	Automated mechanisms exist to detect and alert upon the detection of unauthorized hardware, software and firmware components	5	

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CM-08(03)	System Component Inventory Automated Unauthorized Component Detection	a. Detect the presence of unauthorized hardware, software, and firmware components within the system using [Assignment: organization-defined automated mechanisms] [Assignment: organization-defined frequency]; and b. Take the	Functional	Intersects With	Software Installation Alerts	END-03.1	Mechanisms exist to generate an alert when new software is detected.	5	
CM-08(03)	System Component Inventory Automated Unauthorized Component Detection	a. Detect the presence of unauthorized hardware, software, and firmware components within the system using [Assignment: organization-defined automated mechanisms] [Assignment: organization-defined frequency]; and b. Take the	Functional	Intersects With	Unauthorized Installation Alerts	CFG-05.1	Mechanisms exist to configure systems to generate an alert when the unauthorized installation of software is detected.	5	
CM-09	Configuration Management Plan	Develop, document, and implement a configuration management plan for the system that: a. Addresses roles, responsibilities, and configuration management processes and procedures; b. Establishes a process for identifying	Functional	Subset Of	Configuration Management Program	CFG-01	Mechanisms exist to facilitate the implementation of configuration management controls.	10	
CM-09	Configuration Management Plan	Develop, document, and implement a configuration management plan for the system that: a. Addresses roles, responsibilities, and configuration management processes and procedures; b. Establishes a process for identifying	Functional	Intersects With	Stakeholder Notification of Changes	CHG-05	Mechanisms exist to ensure stakeholders are made aware of and understand the impact of proposed changes.	5	
CM-10	Software Usage Restrictions	a. Use software and associated documentation in accordance with contract agreements and copyright laws; b. Track the use of software and associated documentation protected by	Functional	Equal	Software Usage Restrictions	CFG-04	Mechanisms exist to enforce software usage restrictions to comply with applicable contract agreements and copyright laws.	10	
CM-11	User-installed Software	a. Establish [Assignment: organization-defined policies] governing the installation of software by users; b. Enforce software installation policies through the following methods: [Assignment: organization-defined methods]; and c. Monitor	Functional	Intersects With	User-Installed Software	CFG-05	Mechanisms exist to restrict the ability of non-privileged users to install unauthorized software.	5	
CM-12	Information Location	a. Identify and document the location of [Assignment: organization-defined information] and the specific system components on which the information is processed and stored; b. Identify and document the users who have access to	Functional	Equal	Information Location	DCH-24	Mechanisms exist to identify and document the location of information and the specific system components on which the information resides.	10	
CM-12(01)	Information Location Automated Tools to Support Information Location	Use automated tools to identify [Assignment: organization-defined information by information type] on [Assignment: organization-defined system components] to ensure controls	Functional	Equal	Automated Tools to Support Information Location	DCH-24.1	Automated mechanisms exist to identify by data classification type to ensure adequate security, compliance and resilience controls.	10	
CP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] contingency planning policy that: a. Addresses	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures, at	5	
CP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] contingency planning policy that: a. Addresses	Functional	Subset Of	Business Continuity Management System (BCMS)	BCD-01	Mechanisms exist to facilitate the implementation of contingency planning controls to help ensure	10	
CP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] contingency planning policy that: a. Addresses	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
CP-02	Contingency Plan	a. Develop a contingency plan for the system that: 1. Identifies essential mission and business functions and associated contingency requirements; 2. Provides recovery objectives, restoration priorities, and metrics; 3. Addresses contingency	Functional	Subset Of	Business Continuity Management System (BCMS)	BCD-01	Mechanisms exist to facilitate the implementation of contingency planning controls to help ensure	10	
CP-02	Contingency Plan	a. Develop a contingency plan for the system that: 1. Identifies essential mission and business functions and associated contingency requirements; 2. Provides recovery objectives, restoration priorities, and metrics; 3. Addresses contingency	Functional	Intersects With	Ongoing Contingency Planning	BCD-06	Mechanisms exist to update contingency plans due to changes affecting:	5	
CP-02(01)	Contingency Plan Coordinate with Related Plans	Coordinate contingency plan development with organizational elements responsible for related plans.	Functional	Equal	Coordinate with Related Plans	BCD-01.1	Mechanisms exist to coordinate contingency plan development with internal and external elements responsible for related plans.	10	
CP-02(03)	Contingency Plan Resume Mission and Business Functions	Plan for the resumption of [Selection (one): all; essential] mission and business functions within [Assignment: organization-defined time period] of contingency plan activation.	Functional	Intersects With	Resume All Missions & Business Functions	BCD-02.1	Mechanisms exist to resume all missions and business functions within Recovery Time Objectives (RTOs) of the contingency plan's	5	
CP-02(03)	Contingency Plan Resume Mission and Business Functions	Plan for the resumption of [Selection (one): all; essential] mission and business functions within [Assignment: organization-defined time period] of contingency plan activation.	Functional	Intersects With	Resume Essential Missions & Business Functions	BCD-02.3	Mechanisms exist to resume essential missions and business functions within an organization-defined time period of contingency	5	
CP-02(08)	Contingency Plan Identify Critical Assets	Identify critical system assets supporting [Selection (one): all; essential] mission and business functions.	Functional	Equal	Identify Critical Assets	BCD-02	Mechanisms exist to identify and document the critical Technology Assets, Applications, Services and/or Data (TAAPD) that support essential	10	
CP-03	Contingency Training	a. Provide contingency training to system users consistent with assigned roles and responsibilities: 1. Within [Assignment: organization-defined time period] of assuming a contingency role or responsibility; 2. When required by system changes;	Functional	Equal	Contingency Training	BCD-03	Mechanisms exist to adequately train contingency personnel and applicable stakeholders in their contingency roles and	10	
CP-04	Contingency Plan Testing	a. Test the contingency plan for the system [Assignment: organization-defined frequency] using the following tests to determine the effectiveness of the plan and the readiness to execute the plan; [Assignment: organization-defined tests]; b.	Functional	Intersects With	Contingency Plan Root Cause Analysis (RCA) & Lessons Learned	BCD-05	Mechanisms exist to conduct a Root Cause Analysis (RCA) and "lessons learned" activity every time the contingency plan is activated.	5	
CP-04	Contingency Plan Testing	a. Test the contingency plan for the system [Assignment: organization-defined frequency] using the following tests to determine the effectiveness of the plan and the readiness to execute the plan; [Assignment: organization-defined tests]; b.	Functional	Intersects With	Contingency Plan Testing & Exercises	BCD-04	Mechanisms exist to conduct tests and/or exercises to evaluate the contingency plan's effectiveness and the organization's readiness to	5	
CP-04(01)	Contingency Plan Testing Coordinate with Related Plans	Coordinate contingency plan testing with organizational elements responsible for related plans.	Functional	Equal	Coordinated Testing with Related Plans	BCD-04.1	Mechanisms exist to coordinate contingency plan testing with internal and external elements responsible for related plans.	10	
CP-06	Alternate Storage Site	a. Establish an alternate storage site, including necessary agreements to permit the storage and retrieval of system backup information; and b. Ensure that the alternate storage site provides controls equivalent to that of the primary site.	Functional	Equal	Alternate Storage Site	BCD-08	Mechanisms exist to establish an alternate storage site that includes both the assets and necessary agreements to permit the storage and	10	
CP-06(01)	Alternate Storage Site Separation from Primary Site	Identify an alternate storage site that is sufficiently separated from the primary storage site to reduce susceptibility to the same threats.	Functional	Equal	Separation from Primary Storage Site	BCD-08.1	Mechanisms exist to separate the alternate storage site from the primary storage site to reduce susceptibility to similar threats.	10	
CP-06(03)	Alternate Storage Site Accessibility	Identify potential accessibility problems to the alternate storage site in the event of an area-wide disruption or disaster and outline explicit mitigation actions.	Functional	Equal	Primary Storage Site Accessibility	BCD-08.2	Mechanisms exist to identify and mitigate potential accessibility problems to the alternate storage sites in the event of an area-wide	10	
CP-07	Alternate Processing Site	a. Establish an alternate processing site, including necessary agreements to permit the transfer and resumption of [Assignment: organization-defined system operations] for essential mission and business functions within [Assignment:	Functional	Equal	Alternate Processing Site	BCD-09	Mechanisms exist to establish an alternate processing site that provides security measures equivalent to that of the primary site.	10	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Moderate O**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-mod>

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CP-07(01)	Alternate Processing Site Separation from Primary Site	Identify an alternate processing site that is sufficiently separated from the primary processing site to reduce susceptibility to the same threats.	Functional	Equal	Separation from Primary Processing Site	BCD-09.1	Mechanisms exist to separate the alternate processing site from the primary processing site to reduce susceptibility to similar threats.	10	
CP-07(02)	Alternate Processing Site Accessibility	Identify potential accessibility problems to alternate processing sites in the event of an area-wide disruption or disaster and outlines explicit mitigation actions.	Functional	Equal	Alternate Processing Site Accessibility	BCD-09.2	Mechanisms exist to identify and mitigate potential accessibility problems to the alternate processing sites and possible mitigation actions.	10	
CP-07(03)	Alternate Processing Site Priority of Service	Develop alternate processing site agreements that contain priority-of-service provisions in accordance with availability requirements (including recovery time objectives).	Functional	Equal	Alternate Site Priority of Service	BCD-09.3	Mechanisms exist to address priority-of-service provisions in alternate processing and storage sites that support availability requirements.	10	
CP-08	Telecommunications Services	Establish alternate telecommunications services, including necessary agreements to permit the resumption of [Assignment: organization-defined system operations] for essential mission and business functions within [Assignment: a. Develop primary and alternate telecommunications service agreements that contain priority-of-service provisions in accordance with availability requirements (including recovery time objectives); and b. Request Telecommunications Service	Functional	Intersects With	Telecommunications Services Availability	BCD-10	Mechanisms exist to reduce the likelihood of a single point of failure with primary telecommunications services.	5	
CP-08(01)	Telecommunications Services Priority of Service Provisions	a. Develop primary and alternate telecommunications service agreements that contain priority-of-service provisions in accordance with availability requirements (including recovery time objectives); and b. Request Telecommunications Service	Functional	Equal	Telecommunications Priority of Service Provisions	BCD-10.1	Mechanisms exist to formalize primary and alternate telecommunications service agreements contain priority-of-	10	
CP-08(02)	Telecommunications Services Single Points of Failure	Obtain alternate telecommunications services to reduce the likelihood of sharing a single point of failure with primary telecommunications services.	Functional	Intersects With	Telecommunications Services Availability	BCD-10	Mechanisms exist to reduce the likelihood of a single point of failure with primary telecommunications services.	5	
CP-09	System Backup	a. Conduct backups of user-level information contained in [Assignment: organization-defined system components] [Assignment: organization-defined frequency consistent with recovery time and recovery point objectives]; b. Conduct	Functional	Intersects With	Data Backups	BCD-11	Mechanisms exist to create recurring backups of data, software and/or system images, as well as verify the	5	
CP-09(01)	System Backup Testing for Reliability and Integrity	Test backup information [Assignment: organization-defined frequency] to verify media reliability and information integrity.	Functional	Equal	Testing for Reliability & Integrity	BCD-11.1	Mechanisms exist to routinely test backups that verify the reliability of the backup process, as well as the integrity and availability of the data.	10	
CP-09(08)	System Backup Cryptographic Protection	Implement cryptographic mechanisms to prevent unauthorized disclosure and modification of [Assignment: organization-defined backup information].	Functional	Equal	Cryptographic Protection	BCD-11.4	Cryptographic mechanisms exist to prevent the unauthorized disclosure and/or modification of backup information.	10	
CP-10	System Recovery and Reconstitution	Provide for the recovery and reconstitution of the system to a known state within [Assignment: organization-defined time period consistent with recovery time and recovery point objectives] after a disruption, compromise, or failure.	Functional	Intersects With	Technology Assets, Applications and/or Services (TAAS) Recovery &	BCD-12	Mechanisms exist to ensure the secure recovery and reconstitution of Technology Assets, Applications	5	
CP-10	System Recovery and Reconstitution	Provide for the recovery and reconstitution of the system to a known state within [Assignment: organization-defined time period consistent with recovery time and recovery point objectives] after a disruption, compromise, or failure.	Functional	Intersects With	Business Continuity Management System (BCMS)	BCD-01	Mechanisms exist to facilitate the implementation of contingency planning controls to help ensure	5	
CP-10	System Recovery and Reconstitution	Provide for the recovery and reconstitution of the system to a known state within [Assignment: organization-defined time period consistent with recovery time and recovery point objectives] after a disruption, compromise, or failure.	Functional	Intersects With	Recovery Time / Point Objectives (RTO / RPO)	BCD-01.4	Mechanisms exist to facilitate recovery operations in accordance with Recovery Time Objectives (RTOs) and Recovery Point Objectives	5	
CP-10(02)	System Recovery and Reconstitution Transaction Recovery	Implement transaction recovery for systems that are transaction-based.	Functional	Equal	Transaction Recovery	BCD-12.1	Mechanisms exist to utilize specialized backup mechanisms that will allow transaction recovery for transaction-based Technology	10	
CP-10(06)	System Recovery and Reconstitution Component Protection	Protect system components used for recovery and reconstitution.	Functional	Equal	Backup & Restoration Hardware Protection	BCD-13	Mechanisms exist to protect backup and restoration hardware and software.	10	
CP-12	Safe Mode	When [Assignment: organization-defined conditions] are detected, enter a safe mode of operation with [Assignment: organization-defined restrictions of safe mode of operation].	Functional	Intersects With	Fail Secure	SEA-07.2	Mechanisms exist to enable systems to fail to an organization-defined known-state for types of failures, preserving system state information	5	
IA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] identification and authentication policy that a	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCR), including policies, standards and procedures, at	5	
IA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] identification and authentication policy that a	Functional	Subset Of	Identity & Access Management (IAM)	IAC-01	Mechanisms exist to facilitate the implementation of identification and access management controls.	10	
IA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] identification and authentication policy that a	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
IA-02	Identification and Authentication (organizational Users)	Uniquely identify and authenticate organizational users and associate that unique identification with processes acting on behalf of those users.	Functional	Equal	Identification & Authentication for Organizational Users	IAC-02	Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) organizational users and processes acting on behalf of	10	
IA-02(01)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Multi-Factor Authentication (MFA)	IAC-06	Automated mechanisms exist to enforce Multi-Factor Authentication (MFA) for:	5	
IA-02(01)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Local Access to Privileged Accounts	IAC-06.3	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate local access for privileged accounts.	5	
IA-02(01)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Information Assurance Enabled Products	TDA-02.2	Mechanisms exist to limit the use of commercially-provided Information Assurance (IA) and IA-enabled IT products to those products that have	5	
IA-02(01)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Out-of-Band Multi-Factor Authentication	IAC-06.4	Mechanisms exist to implement Multi-Factor Authentication (MFA) for access to privileged and non-privileged accounts such that one of	5	
IA-02(01)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Network Access to Privileged Accounts	IAC-06.1	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate network access for privileged accounts.	5	
IA-02(01)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Network Access to Non-Privileged Accounts	IAC-06.2	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate network access for non-privileged accounts.	5	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Moderate O**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-mod>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
IA-02(01)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Hardware Token-Based Authentication	IAC-10.7	Automated mechanisms exist to ensure organization-defined token quality requirements are satisfied for hardware token-based	5	
IA-02(02)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Information Assurance Enabled Products	TDA-02.2	Mechanisms exist to limit the use of commercially-provided Information Assurance (IA) and IA-enabled IT products to those products that have	5	
IA-02(02)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Network Access to Non-Privileged Accounts	IAC-06.2	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate network access for non-privileged accounts.	5	
IA-02(02)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Out-of-Band Multi-Factor Authentication	IAC-06.4	Mechanisms exist to implement Multi-Factor Authentication (MFA) for access to privileged and non-privileged accounts such that one of	5	
IA-02(02)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Hardware Token-Based Authentication	IAC-10.7	Automated mechanisms exist to ensure organization-defined token quality requirements are satisfied for hardware token-based	5	
IA-02(02)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Network Access to Privileged Accounts	IAC-06.1	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate network access for privileged accounts.	5	
IA-02(02)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Multi-Factor Authentication (MFA)	IAC-06	Automated mechanisms exist to enforce Multi-Factor Authentication (MFA) for:	5	
IA-02(02)	Identification and Authentication (organizational Users) Multi-factor	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Local Access to Privileged Accounts	IAC-06.3	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate local access for privileged accounts.	5	
IA-02(08)	Identification and Authentication (organizational Users) Access to Accounts	Implement replay-resistant authentication mechanisms for access to [Selection (one or more): privileged accounts; non-privileged accounts].	Functional	Equal	Replay-Resistant Authentication	IAC-02.2	Automated mechanisms exist to employ replay-resistant authentication.	10	
IA-02(12)	Identification and Authentication (organizational Users) Acceptance of PIV	Accept and electronically verify Personal Identity Verification-compliant credentials.	Functional	Intersects With	Acceptance of PIV Credentials	IAC-02.3	Mechanisms exist to accept and electronically verify organizational Personal Identity Verification (PIV) credentials.	5	
IA-03	Device Identification and Authentication	Uniquely identify and authenticate [Assignment: organization-defined devices and/or types of devices] before establishing a [Selection (one or more): local; remote; network] connection.	Functional	Intersects With	Identification & Authentication for Devices	IAC-04	Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) devices before establishing a connection using	5	
IA-04	Identifier Management	Manage system identifiers by:a. Receiving authorization from [Assignment: organization-defined personnel or roles] to assign an individual, group, role, service, or device identifier;b. Selecting an identifier that identifies an individual, group, role	Functional	Intersects With	Authenticate, Authorize and Audit (AAA)	IAC-01.2	Mechanisms exist to strictly govern the use of Authenticate, Authorize and Audit (AAA) solutions, both on-premises and those hosted by an	5	
IA-04	Identifier Management	Manage system identifiers by:a. Receiving authorization from [Assignment: organization-defined personnel or roles] to assign an individual, group, role, service, or device identifier;b. Selecting an identifier that identifies an individual, group, role	Functional	Intersects With	Identifier Management (User Names)	IAC-09	Mechanisms exist to govern naming standards for usernames and Technology Assets, Applications and/or Services (TAAS)	5	
IA-04(04)	Identifier Management Identify User Status	Manage individual identifiers by uniquely identifying each individual as [Assignment: organization-defined characteristic identifying individual status].	Functional	Intersects With	Authenticate, Authorize and Audit (AAA)	IAC-01.2	Mechanisms exist to strictly govern the use of Authenticate, Authorize and Audit (AAA) solutions, both on-premises and those hosted by an	5	
IA-04(04)	Identifier Management Identify User Status	Manage individual identifiers by uniquely identifying each individual as [Assignment: organization-defined characteristic identifying individual status].	Functional	Intersects With	User Identity (ID) Management	IAC-09.1	Mechanisms exist to ensure proper user identification management for non-consumer users and administrators	5	
IA-04(04)	Identifier Management Identify User Status	Manage individual identifiers by uniquely identifying each individual as [Assignment: organization-defined characteristic identifying individual status].	Functional	Intersects With	Identity User Status	IAC-09.2	Mechanisms exist to identify contractors and other third-party users through unique username characteristics	5	
IA-05	Authenticator Management	Manage system authenticators by:a. Verifying, as part of the initial authenticator distribution, the identity of the individual, group, role, service, or device receiving the authenticator;b. Establishing initial authenticator content for any authenticators	Functional	Intersects With	Authenticator Management	IAC-10	Mechanisms exist to: (1) Securely manage authenticators for users and devices; and (2) Ensure the strength of	5	
IA-05	Authenticator Management	Manage system authenticators by:a. Verifying, as part of the initial authenticator distribution, the identity of the individual, group, role, service, or device receiving the authenticator;b. Establishing initial authenticator content for any authenticators	Functional	Intersects With	Default Authenticators	IAC-10.8	Mechanisms exist to ensure default authenticators are changed as part of account creation or system installation	5	
IA-05(01)	Authenticator Management Password-based Authentication	For password-based authentication:a. Maintain a list of commonly-used, expected, or compromised passwords and update the list [Assignment: organization-defined frequency] and when organizational passwords are suspected to have	Functional	Intersects With	Automated Support For Password Strength	IAC-10.4	Automated mechanisms exist to determine if password authenticators are sufficiently strong enough to satisfy organization-defined password	5	
IA-05(01)	Authenticator Management Password-based Authentication	For password-based authentication:a. Maintain a list of commonly-used, expected, or compromised passwords and update the list [Assignment: organization-defined frequency] and when organizational passwords are suspected to have	Functional	Intersects With	Password-Based Authentication	IAC-10.1	Mechanisms exist to enforce complexity, length and lifespan considerations to ensure strong criteria for password-based	5	
IA-05(01)	Authenticator Management Password-based Authentication	For password-based authentication:a. Maintain a list of commonly-used, expected, or compromised passwords and update the list [Assignment: organization-defined frequency] and when organizational passwords are suspected to have	Functional	Intersects With	Authenticator Management	IAC-10	Mechanisms exist to: (1) Securely manage authenticators for users and devices; and (2) Ensure the strength of	5	
IA-05(02)	Management Public Key-based Authentication	a. For public key-based authentication:1. Enforce authorized access to the corresponding private key; and2. Map the authenticated identity to the account of the individual or group; andb. When public key infrastructure (PKI) is used:1. Validate	Functional	Equal	PKI-Based Authentication	IAC-10.2	Automated mechanisms exist to validate certificates by constructing and verifying a certification path to an accepted trust anchor including	10	
IA-05(06)	Authenticator Management Protection of Authenticators	Protect authenticators commensurate with the security category of the information to which use of the authenticator permits access.	Functional	Intersects With	User Responsibilities for Account Management	IAC-18	Mechanisms exist to compel users to follow accepted practices in the use of authentication mechanisms (e.g., passwords, passphrases, physical or	5	
IA-05(06)	Authenticator Management Protection of Authenticators	Protect authenticators commensurate with the security category of the information to which use of the authenticator permits access.	Functional	Intersects With	Protection of Authenticators	IAC-10.5	Mechanisms exist to protect authenticators commensurate with the sensitivity of the information to which use of the authenticator	5	
IA-06	Authentication Feedback	Obscure feedback of authentication information during the authentication process to protect the information from possible exploitation and use by unauthorized individuals.	Functional	Equal	Authenticator Feedback	IAC-11	Mechanisms exist to obscure the feedback of authentication information during the authentication process to protect the information	10	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Moderate O**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-mod>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
IA-07	Cryptographic Module Authentication	Implement mechanisms for authentication to a cryptographic module that meet the requirements of applicable laws, executive orders, directives, policies, regulations, standards, and guidelines for such authentication.	Functional	Intersects With	Cryptographic Module Authentication	IAC-12	Mechanisms exist to ensure cryptographic modules adhere to applicable statutory, regulatory and contractual requirements for security.	5	
IA-07	Cryptographic Module Authentication	Implement mechanisms for authentication to a cryptographic module that meet the requirements of applicable laws, executive orders, directives, policies, regulations, standards, and guidelines for such authentication.	Functional	Intersects With	Automated Authentication Through Cryptographic	CRY-02	Automated mechanisms exist to enable systems to authenticate to a cryptographic module.	5	
IA-08	Identification and Authentication (non-organizational Users)	Uniquely identify and authenticate non-organizational users or processes acting on behalf of non-organizational users.	Functional	Equal	Identification & Authentication for Non-Organizational Users	IAC-03	Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) third-party users and processes that provide services to the	10	
IA-08(01)	Identification and Authentication (non-organizational Users) Acceptance of PIV	Accept and electronically verify Personal Identity Verification-compliant credentials from other federal agencies.	Functional	Equal	Acceptance of PIV Credentials from Other Organizations	IAC-03.1	Mechanisms exist to accept and electronically verify Personal Identity Verification (PIV) credentials from third-parties	10	
IA-08(02)	Identification and Authentication (non-organizational Users) Acceptance of	a. Accept only external authenticators that are NIST-compliant; andb. Document and maintain a list of accepted external authenticators.	Functional	Equal	Acceptance of Third-Party Credentials	IAC-03.2	Automated mechanisms exist to accept Federal Identity, Credential and Access Management (FICAM)-approved third-party credentials.	10	
IA-08(04)	Identification and Authentication (non-organizational Users) Use of Defined	Conform to the following profiles for identity management [Assignment: organization-defined identity management profiles].	Functional	Equal	Use of FICAM-Issued Profiles	IAC-03.3	Mechanisms exist to conform systems to Federal Identity, Credential and Access Management (FICAM)-issued profiles.	10	
IA-11	Re-authentication	Require users to re-authenticate when [Assignment: organization-defined circumstances or situations requiring re-authentication].	Functional	Equal	Re-Authentication	IAC-14	Mechanisms exist to force users and devices to re-authenticate according to organization-defined circumstances that necessitate re-	10	
IA-12	Identity Proofing	a. Identity proof users that require accounts for logical access to systems based on appropriate identity assurance level requirements as specified in applicable standards and guidelines;b. Resolve user identities to a unique individual;	Functional	Equal	Identity Proofing (Identity Verification)	IAC-28	Mechanisms exist to verify the identity of a user before issuing authenticators or modifying access permissions.	10	
IA-12(02)	Identity Proofing Identity Evidence	Require evidence of individual identification be presented to the registration authority.	Functional	Equal	Identity Evidence	IAC-28.2	Mechanisms exist to require evidence of individual identification to be presented to the registration authority.	10	
IA-12(03)	Identity Proofing Identity Evidence Validation and Verification	Require that the presented identity evidence be validated and verified through [Assignment: organizational defined methods of validation and verification].	Functional	Equal	Identity Evidence Validation & Verification	IAC-28.3	Mechanisms exist to require that the presented identity evidence be validated and verified through organizational-defined methods of	10	
IA-12(05)	Identity Proofing Address Confirmation	Require that a [Selection (one): registration code; notice of proofing] be delivered through an out-of-band channel to verify the users address (physical or digital) of record.	Functional	Equal	Address Confirmation	IAC-28.5	Mechanisms exist to require that a notice of proofing be delivered through an out-of-band channel to verify the user's address (physical or	10	
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] incident response policy that a. Addresses	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] incident response policy that a. Addresses	Functional	Subset Of	Incident Response Operations	IRO-01	Mechanisms exist to implement and govern processes and documentation to facilitate an organization-wide response capability for cybersecurity	10	
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] incident response policy that a. Addresses	Functional	Intersects With	IRP Update	IRO-04.2	Mechanisms exist to regularly review and modify incident response practices to incorporate lessons learned, business process changes	5	
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] incident response policy that a. Addresses	Functional	Intersects With	Root Cause Analysis (RCA) & Lessons Learned	IRO-13	Mechanisms exist to incorporate lessons learned from analyzing and resolving cybersecurity and data protection incidents to reduce the	5	
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] incident response policy that a. Addresses	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures, at	5	
IR-02	Incident Response Training	a. Provide incident response training to system users consistent with assigned roles and responsibilities;1. Within [Assignment: organization-defined time period] of assuming an incident response role or responsibility or acquiring system	Functional	Intersects With	Incident Response Training	IRO-05	Mechanisms exist to train personnel in their incident response roles and responsibilities.	5	
IR-03	Incident Response Testing	Test the effectiveness of the incident response capability for the system [Assignment: organization-defined frequency] using the following tests: [Assignment: organization-defined tests].	Functional	Intersects With	Incident Response Testing	IRO-06	Mechanisms exist to formally test incident response capabilities through realistic exercises to determine the operational	5	
IR-03(02)	Incident Response Testing Coordination with Related Plans	Coordinate incident response testing with organizational elements responsible for related plans.	Functional	Equal	Coordination with Related Plans	IRO-06.1	Mechanisms exist to coordinate incident response testing with organizational elements responsible for related plans.	10	
IR-04	Incident Handling	a. Implement an incident handling capability for incidents that is consistent with the incident response plan and includes preparation, detection and analysis, containment, eradication, and recovery;b. Coordinate incident handling activities with	Functional	Equal	Incident Handling	IRO-02	Mechanisms exist to cover: (1) Preparation; (2) Automated event detection or manual incident report intake;	10	
IR-04(01)	Incident Handling Automated Incident Handling Processes	Support the incident handling process using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Automated Incident Handling Processes	IRO-02.1	Automated mechanisms exist to support the incident handling process.	10	
IR-05	Incident Monitoring	Track and document incidents.	Functional	Equal	Situational Awareness For Incidents	IRO-09	Mechanisms exist to document, monitor and report the status of cybersecurity and data protection incidents to internal stakeholders all	10	
IR-06	Incident Reporting	a. Require personnel to report suspected incidents to the organizational incident response capability within [Assignment: organization-defined time period]; andb. Report incident information to [Assignment: organization-defined authorities];	Functional	Intersects With	Incident Stakeholder Reporting	IRO-10	Mechanisms exist to timely-report incidents to applicable: (1) Internal stakeholders; (2) Affected clients & third-parties;	5	
IR-06	Incident Reporting	a. Require personnel to report suspected incidents to the organizational incident response capability within [Assignment: organization-defined time period]; andb. Report incident information to [Assignment: organization-defined authorities];	Functional	Intersects With	Regulatory & Law Enforcement Contacts	IRO-14	Mechanisms exist to maintain incident response contacts with applicable regulatory and law enforcement agencies.	5	
IR-06	Incident Reporting	a. Require personnel to report suspected incidents to the organizational incident response capability within [Assignment: organization-defined time period]; andb. Report incident information to [Assignment: organization-defined authorities];	Functional	Intersects With	Contacts With Authorities	GOV-06	Mechanisms exist to identify and document appropriate contacts with relevant law enforcement and regulatory bodies.	5	

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Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Moderate O**

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FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
IR-06(01)	Incident Reporting Automated Reporting	Report incidents using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Automated Reporting	IRO-10.1	Automated mechanisms exist to assist in the reporting of cybersecurity and data protection incidents.	10	
IR-06(03)	Incident Reporting Supply Chain Coordination	Provide incident information to the provider of the product or service and other organizations involved in the supply chain or supply chain governance for systems or system components <u>related to the incident</u> .	Functional	Intersects With	Supply Chain Coordination	IRO-10.4	Mechanisms exist to provide cybersecurity and data protection incident information to the provider of the Technology Assets, Applications, and Services (TAAS).	5	
IR-07	Incident Response Assistance	Provide an incident response support resource, integral to the organizational incident response capability, that offers advice and assistance to users of the system for the handling and <u>reporting of incidents</u> .	Functional	Equal	Incident Reporting Assistance	IRO-11	Mechanisms exist to provide incident response advice and assistance to users of Technology Assets, Applications, and Services (TAAS).	10	
IR-07(01)	Incident Response Assistance Automation Support for Availability of	Increase the availability of incident response information and support using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Automation Support of Availability of Information / Support	IRO-11.1	Automated mechanisms exist to increase the availability of incident response-related information and support.	10	
IR-08	Incident Response Plan	a. Develop an incident response plan that:1. Provides the organization with a roadmap for implementing its incident response capability;2. Describes the structure and <u>organization of the incident response capability</u> ;3. <u>Provides a</u>	Functional	Equal	Incident Response Plan (IRP)	IRO-04	Mechanisms exist to maintain and make available a current and viable Incident Response Plan (IRP) to all stakeholders.	10	
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; <u>System-level</u>] <u>maintenance policy that a. Addresses purpose</u>	Functional	Subset Of	Maintenance Operations	MNT-01	Mechanisms exist to develop, disseminate, review & update procedures to facilitate the <u>implementation of maintenance</u>	10	
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; <u>System-level</u>] <u>maintenance policy that a. Addresses purpose</u>	Functional	Intersects With	Remote Maintenance Notifications	MNT-05.2	Mechanisms exist to require maintenance personnel to notify affected stakeholders when remote, <u>non-local maintenance is planned</u>	5	
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; <u>System-level</u>] <u>maintenance policy that a. Addresses purpose</u>	Functional	Intersects With	Auditing Remote Maintenance	MNT-05.1	Mechanisms exist to audit remote, non-local maintenance and diagnostic sessions, as well as review <u>the maintenance action performed</u>	5	
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; <u>System-level</u>] <u>maintenance policy that a. Addresses purpose</u>	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, <u>standards and procedures at</u>	5	
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; <u>System-level</u>] <u>maintenance policy that a. Addresses purpose</u>	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary <u>for secure, compliant and resilient</u>	5	
MA-02	Controlled Maintenance	a. Schedule, document, and review records of maintenance, repair, and replacement on system components in accordance with manufacturer or vendor specifications and/or <u>organizational requirements</u> ;b. <u>Approve and monitor all</u>	Functional	Equal	Controlled Maintenance	MNT-02	Mechanisms exist to conduct controlled maintenance activities throughout the lifecycle of the Technology Assets, Applications, and Services (TAAS).	10	
MA-03	Maintenance Tools	a. Approve, control, and monitor the use of system maintenance tools; andb. Review previously approved system maintenance tools [Assignment: organization-defined frequency].	Functional	Intersects With	Maintenance Tools	MNT-04	Mechanisms exist to control and monitor the use of system maintenance tools.	5	
MA-03(01)	Maintenance Tools Inspect Tools	Inspect the maintenance tools used by maintenance personnel for improper or unauthorized modifications.	Functional	Equal	Inspect Tools	MNT-04.1	Mechanisms exist to inspect maintenance tools carried into a facility by maintenance personnel for <u>improper or unauthorized</u>	10	
MA-03(02)	Maintenance Tools Inspect Media	Check media containing diagnostic and test programs for malicious code before the media are used in the system.	Functional	Equal	Inspect Media	MNT-04.2	Mechanisms exist to check media containing diagnostic and test programs for malicious code before <u>the media are used</u>	10	
MA-03(03)	Maintenance Tools Prevent Unauthorized Removal	Prevent the removal of maintenance equipment containing organizational information by:a. Verifying that there is no organizational information contained on the equipment;b. <u>Sanitizing or destroying the equipment</u> ;c. <u>Retaining the</u>	Functional	Equal	Prevent Unauthorized Removal	MNT-04.3	Mechanisms exist to prevent or control the removal of equipment undergoing maintenance that <u>contains organizational information</u>	10	
MA-04	Nonlocal Maintenance	a. Approve and monitor nonlocal maintenance and diagnostic activities;b. Allow the use of nonlocal maintenance and diagnostic tools only as consistent with organizational policy <u>and documented in the security plan for the system</u> ;c. <u>Employ</u>	Functional	Intersects With	Remote Maintenance	MNT-05	Mechanisms exist to authorize, monitor and control remote, non-local maintenance and diagnostic <u>activities</u>	5	
MA-04	Nonlocal Maintenance	a. Approve and monitor nonlocal maintenance and diagnostic activities;b. Allow the use of nonlocal maintenance and diagnostic tools only as consistent with organizational policy <u>and documented in the security plan for the system</u> ;c. <u>Employ</u>	Functional	Intersects With	Remote Maintenance Notifications	MNT-05.2	Mechanisms exist to require maintenance personnel to notify affected stakeholders when remote, <u>non-local maintenance is planned</u>	5	
MA-04	Nonlocal Maintenance	a. Approve and monitor nonlocal maintenance and diagnostic activities;b. Allow the use of nonlocal maintenance and diagnostic tools only as consistent with organizational policy <u>and documented in the security plan for the system</u> ;c. <u>Employ</u>	Functional	Intersects With	Auditing Remote Maintenance	MNT-05.1	Mechanisms exist to audit remote, non-local maintenance and diagnostic sessions, as well as review <u>the maintenance action performed</u>	5	
MA-04(01)	Nonlocal Maintenance Logging and Review	a. Log [Assignment: organization-defined audit events] for nonlocal maintenance and diagnostic sessions; andb. Review the audit records of the maintenance and diagnostic sessions <u>to detect anomalous behavior</u>	Functional	Intersects With	Auditing Remote Maintenance	MNT-05.1	Mechanisms exist to audit remote, non-local maintenance and diagnostic sessions, as well as review <u>the maintenance action performed</u>	5	
MA-05	Maintenance Personnel	a. Establish a process for maintenance personnel authorization and maintain a list of authorized maintenance organizations or personnel;b. Verify that non-escorted <u>personnel performing maintenance on the system possess the</u>	Functional	Equal	Authorized Maintenance Personnel	MNT-06	Mechanisms exist to maintain a current list of authorized maintenance organizations or <u>personnel</u>	10	
MA-06	Timely Maintenance	Obtain maintenance support and/or spare parts for [Assignment: organization-defined system components] within [Assignment: organization-defined time period] of failure.	Functional	Equal	Timely Maintenance	MNT-03	Mechanisms exist to obtain maintenance support and/or spare parts for Technology Assets, Applications, and Services (TAAS).	10	
MA-07	Field Maintenance	Restrict or prohibit field maintenance on [Assignment: organization-defined systems or system components] to [Assignment: organization-defined trusted maintenance facilities].	Functional	Equal	Field Maintenance	MNT-08	Mechanisms exist to securely conduct field maintenance on geographically deployed assets.	10	
MP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; <u>System-level</u>] <u>media protection policy that a. Addresses</u>	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, <u>standards and procedures at</u>	5	
MP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; <u>System-level</u>] <u>media protection policy that a. Addresses</u>	Functional	Subset Of	Data Protection	DCH-01	Mechanisms exist to facilitate the implementation of data protection controls.	10	
MP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; <u>System-level</u>] <u>media protection policy that a. Addresses</u>	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary <u>for secure, compliant and resilient</u>	5	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Moderate O**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-mod>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
MP-02	Media Access	Restrict access to [Assignment: organization-defined types of digital and/or non-digital media] to [Assignment: organization-defined personnel or roles].	Functional	Intersects With	Media Access	DCH-03	Mechanisms exist to control and restrict access to digital and non-digital media to authorized individuals.	5	
MP-02	Media Access	Restrict access to [Assignment: organization-defined types of digital and/or non-digital media] to [Assignment: organization-defined personnel or roles].	Functional	Intersects With	Endpoint Device Management (EDM)	END-01	Mechanisms exist to facilitate the implementation of Endpoint Device Management (EDM) controls.	5	
MP-03	Media Marking	a. Mark system media indicating the distribution limitations, handling caveats, and applicable security markings (if any) of the information; andb. Exempt [Assignment: organization-defined types of system media] from marking if the media	Functional	Intersects With	Media Marking	DCH-04	Mechanisms exist to mark media in accordance with data protection requirements so that personnel are alerted to distribution limitations.	5	
MP-03	Media Marking	a. Mark system media indicating the distribution limitations, handling caveats, and applicable security markings (if any) of the information; andb. Exempt [Assignment: organization-defined types of system media] from marking if the media	Functional	Intersects With	Automated Marking	DCH-04.1	Automated mechanisms exist to mark physical media and digital files to indicate the distribution limitations, handling requirements and applicable	5	
MP-04	Media Storage	a. Physically control and securely store [Assignment: organization-defined types of digital and/or non-digital media] within [Assignment: organization-defined controlled areas]; andb. Protect system media types defined in MP-04a until the	Functional	Equal	Media Storage	DCH-06	Mechanisms exist to: (1) Physically control and securely store digital and non-digital media within controlled areas using	10	
MP-05	Media Transport	a. Protect and control [Assignment: organization-defined types of system media] during transport outside of controlled areas using [Assignment: organization-defined controls];b. Maintain accountability for system media during transport outside of	Functional	Equal	Media Transportation	DCH-07	Mechanisms exist to protect and control digital and non-digital media during transport outside of controlled areas using appropriate security	10	
MP-06	Media Sanitization	a. Sanitize [Assignment: organization-defined system media] prior to disposal, release out of organizational control, or release for reuse using [Assignment: organization-defined sanitization techniques and procedures]; andb. Employ	Functional	Intersects With	Physical Media Disposal	DCH-08	Mechanisms exist to securely dispose of media when it is no longer required, using formal procedures.	5	
MP-06	Media Sanitization	a. Sanitize [Assignment: organization-defined system media] prior to disposal, release out of organizational control, or release for reuse using [Assignment: organization-defined sanitization techniques and procedures]; andb. Employ	Functional	Intersects With	System Media Sanitization	DCH-09	Mechanisms exist to sanitize system media with the strength and integrity commensurate with the classification or sensitivity of the information prior	5	
MP-06	Media Sanitization	a. Sanitize [Assignment: organization-defined system media] prior to disposal, release out of organizational control, or release for reuse using [Assignment: organization-defined sanitization techniques and procedures]; andb. Employ	Functional	Intersects With	Sanitization of Personal Data (PD)	DCH-09.3	Mechanisms exist to facilitate the sanitization of Personal Data (PD).	5	
MP-07	Media Use	a. [Selection (one): Restrict; Prohibit] the use of [Assignment: organization-defined types of system media] on [Assignment: organization-defined systems or system components] using [Assignment: organization-defined controls]; andb. Prohibit the	Functional	Intersects With	Media & Data Retention	DCH-18	Mechanisms exist to retain media and data in accordance with applicable statutory, regulatory and contractual obligations.	5	
MP-07	Media Use	a. [Selection (one): Restrict; Prohibit] the use of [Assignment: organization-defined types of system media] on [Assignment: organization-defined systems or system components] using [Assignment: organization-defined controls]; andb. Prohibit the	Functional	Intersects With	Media Use	DCH-10	Mechanisms exist to restrict the use of types of digital media on systems or system components.	5	
MP-07	Media Use	a. [Selection (one): Restrict; Prohibit] the use of [Assignment: organization-defined types of system media] on [Assignment: organization-defined systems or system components] using [Assignment: organization-defined controls]; andb. Prohibit the	Functional	Intersects With	Prohibit Use Without Owner	DCH-10.2	Mechanisms exist to prohibit the use of portable storage devices in organizational systems when such	5	
PE-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] physical and environmental protection policy	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
PE-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] physical and environmental protection policy	Functional	Subset Of	Physical & Environmental Protections	PES-01	Mechanisms exist to facilitate the operation of physical and environmental protection controls.	10	
PE-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] physical and environmental protection policy	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies,	5	
PE-02	Physical Access Authorizations	a. Develop, approve, and maintain a list of individuals with authorized access to the facility where the system resides;b. Issue authorization credentials for facility access;c. Review the access list detailing authorized facility access by individuals	Functional	Equal	Physical Access Authorizations	PES-02	Physical access control mechanisms exist to maintain a current list of personnel with authorized access to organizational facilities (except for	10	
PE-03	Physical Access Control	a. Enforce physical access authorizations at [Assignment: organization-defined entry and exit points to the facility where the system resides] by;1. Verifying individual access authorizations before granting access to the facility; and2.	Functional	Intersects With	Physical Access Control	PES-03	Physical access control mechanisms exist to enforce physical access authorizations for all physical access points (including designated	5	
PE-04	Access Control for Transmission	Control physical access to [Assignment: organization-defined system distribution and transmission lines] within organizational facilities using [Assignment: organization-defined security controls].	Functional	Equal	Transmission Medium Security	PES-12.1	Physical security mechanisms exist to protect power and telecommunications cabling carrying data or supporting information	10	
PE-05	Access Control for Output Devices	Control physical access to output from [Assignment: organization-defined output devices] to prevent unauthorized individuals from obtaining the output.	Functional	Equal	Access Control for Output Devices	PES-12.2	Physical security mechanisms exist to restrict access to printers and other system output devices to prevent unauthorized individuals from	10	
PE-06	Monitoring Physical Access	a. Monitor physical access to the facility where the system resides to detect and respond to physical security incidents;b. Review physical access logs [Assignment: organization-defined frequency] and upon occurrence of [Assignment: organization-	Functional	Equal	Monitoring Physical Access	PES-05	Physical access control mechanisms exist to monitor for, detect and respond to physical security incidents	10	
PE-06(01)	Monitoring Physical Access Intrusion Alarms and Surveillance	Monitor physical access to the facility where the system resides using physical intrusion alarms and surveillance equipment.	Functional	Equal	Intrusion Alarms / Surveillance Equipment	PES-05.1	Physical access control mechanisms exist to monitor physical intrusion alarms and surveillance equipment.	10	
PE-06(04)	Monitoring Physical Access Monitoring Physical Access to Systems	Monitor physical access to the system in addition to the physical access monitoring of the facility at [Assignment: organization-defined physical spaces containing one or more components of the system]	Functional	Equal	Monitoring Physical Access To Critical Systems	PES-05.2	Facility security mechanisms exist to monitor physical access to critical systems or sensitive and/or regulated data in addition to the physical	10	
PE-08	Visitor Access Records	a. Maintain visitor access records to the facility where the system resides for [Assignment: organization-defined time period];b. Review visitor access records [Assignment: organization-defined frequency]; andc. Report anomalies in	Functional	Equal	Physical Access Logs	PES-03.3	Physical access control mechanisms generate a log entry for each access attempt through controlled ingress and egress points.	10	
PE-09	Power Equipment and Cabling	Protect power equipment and power cabling for the system from damage and destruction.	Functional	Equal	Supporting Utilities	PES-07	Facility security mechanisms exist to protect power equipment and power cabling for the system from damage and destruction.	10	
PE-10	Emergency Shutoff	a. Provide the capability of shutting off power to [Assignment: organization-defined system or individual system components] in emergency situations;b. Place emergency shutoff switches or devices in [Assignment: organization-defined location] by	Functional	Equal	Emergency Shutoff	PES-07.2	Facility security mechanisms exist to shut off power in emergency situations by: (1) Placing emergency shutoff	10	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Moderate O**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-mod>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
PE-11	Emergency Power	Provide an uninterruptible power supply to facilitate [Selection (one or more): an orderly shutdown of the system; transition of the system to long-term alternate power] in the event of a <u>primary power source loss</u> .	Functional	Intersects With	Emergency Power	PES-07.3	Facility security mechanisms exist to supply alternate power, capable of maintaining minimally-required operational capability in the event of	5	
PE-12	Emergency Lighting	Employ and maintain automatic emergency lighting for the system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility.	Functional	Equal	Emergency Lighting	PES-07.4	Facility security mechanisms exist to utilize and maintain automatic emergency lighting that activates in the event of a power outage or	10	
PE-13	Fire Protection	Employ and maintain fire detection and suppression systems that are supported by an independent energy source.	Functional	Equal	Fire Protection	PES-08	Facility security mechanisms exist to utilize and maintain fire suppression and detection devices/systems for the system that are supported by an	10	
PE-13(01)	Fire Protection Detection Systems — Automatic Activation and Notification	Employ fire detection systems that activate automatically and notify [Assignment: organization-defined personnel or roles] and [Assignment: organization-defined emergency responders] in the event of a fire.	Functional	Equal	Fire Detection Devices	PES-08.1	Facility security mechanisms exist to utilize and maintain fire detection devices/systems that activate automatically and notify	10	
PE-14	Environmental Controls	a. Maintain [Selection (one or more): temperature; humidity; pressure; radiation; [Assignment: organization-defined environmental control]] levels within the facility where the system resides at [Assignment: organization-defined	Functional	Equal	Temperature & Humidity Controls	PES-09	Facility security mechanisms exist to maintain and monitor temperature and humidity levels within the facility.	10	
PE-15	Water Damage Protection	Protect the system from damage resulting from water leakage by providing master shutoff or isolation valves that are accessible, working properly, and known to key personnel.	Functional	Equal	Water Damage Protection	PES-07.5	Facility security mechanisms exist to protect systems from damage resulting from water leakage by providing master shutoff valves that	10	
PE-16	Delivery and Removal	a. Authorize and control [Assignment: organization-defined types of system components] entering and exiting the facility; and b. Maintain records of the system components.	Functional	Equal	Delivery & Removal	PES-10	Physical security mechanisms exist to isolate information processing facilities from points such as delivery and loading areas and other points to	10	
PE-17	Alternate Work Site	a. Determine and document the [Assignment: organization-defined alternate work sites] allowed for use by employees; b. Employ the following controls at alternate work sites: [Assignment: organization-defined controls]; c. Assess the Mark [Assignment: organization-defined system hardware components] indicating the impact level or classification level of the information permitted to be processed, stored, or transmitted by the hardware component.	Functional	Equal	Alternate Work Site	PES-11	Physical security mechanisms exist to utilize appropriate management, operational and technical controls at alternate work sites.	10	
PE-22	Component Marking	Mark [Assignment: organization-defined system hardware components] indicating the impact level or classification level of the information permitted to be processed, stored, or transmitted by the hardware component.	Functional	Intersects With	Asset Scope Classification	AST-04.1	Mechanisms exist to determine security, compliance and resilience control applicability by identifying, assessing and documenting the	5	
PE-22	Component Marking	Mark [Assignment: organization-defined system hardware components] indicating the impact level or classification level of the information permitted to be processed, stored, or transmitted by the hardware component.	Functional	Intersects With	Component Marking	PES-16	Physical security mechanisms exist to mark system hardware components indicating the impact or classification level of the information	5	
PL-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] planning policy that: a. Addresses purpose	Functional	Subset Of	Statutory, Regulatory & Contractual Compliance	CPL-01	Mechanisms exist to facilitate the identification and implementation of relevant statutory, regulatory and contractual controls.	10	
PL-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] planning policy that: a. Addresses purpose	Functional	Subset Of	Technology Development & Acquisition	TDA-01	Mechanisms exist to facilitate the implementation of tailored development and acquisition strategies, contract tools and	10	
PL-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] planning policy that: a. Addresses purpose	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures at	5	
PL-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] planning policy that: a. Addresses purpose	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
PL-02	System Security and Privacy Plans	a. Develop security and privacy plans for the system that: 1. Are consistent with the organization's enterprise architecture; 2. Explicitly define the constituent system components; 3. Describe the operational context of the system in terms of	Functional	Intersects With	Plan / Coordinate with Other Organizational Entities	IAO-03.1	Mechanisms exist to plan and coordinate Information Assurance Program (IAP) activities with affected stakeholders before conducting such	5	
PL-02	System Security and Privacy Plans	a. Develop security and privacy plans for the system that: 1. Are consistent with the organization's enterprise architecture; 2. Explicitly define the constituent system components; 3. Describe the operational context of the system in terms of	Functional	Intersects With	Applied Security, Compliance and Resilience Controls Documentation	IAO-03	Mechanisms exist to generate authoritative documentation (e.g., System Security Plan (SSP)) that: (1) Identify key architectural and	5	
PL-02	System Security and Privacy Plans	a. Develop security and privacy plans for the system that: 1. Are consistent with the organization's enterprise architecture; 2. Explicitly define the constituent system components; 3. Describe the operational context of the system in terms of	Functional	Intersects With	Network Diagrams & Data Flow Diagrams (DFDs)	AST-04	Mechanisms exist to maintain network architecture diagrams that: (1) Contain sufficient detail to assess the security of the network's	5	
PL-04	Rules of Behavior	a. Establish and provide to individuals requiring access to the system, the rules that describe their responsibilities and expected behavior for information and system usage, security, and privacy; b. Receive a documented acknowledgment from	Functional	Intersects With	Terms of Employment	HRS-05	Mechanisms exist to require all employees and contractors to apply cybersecurity and data protection principles in their daily work to enable	5	
PL-04	Rules of Behavior	a. Establish and provide to individuals requiring access to the system, the rules that describe their responsibilities and expected behavior for information and system usage, security, and privacy; c. Receive a documented acknowledgment from	Functional	Intersects With	Rules of Behavior	HRS-05.1	Mechanisms exist to define acceptable and unacceptable rules of behavior for the use of technologies, including consequences for	5	
PL-04	Rules of Behavior	a. Establish and provide to individuals requiring access to the system, the rules that describe their responsibilities and expected behavior for information and system usage, security, and privacy; c. Receive a documented acknowledgment from	Functional	Intersects With	Technology Use Restrictions	HRS-05.3	Mechanisms exist to establish usage restrictions and implementation guidance for organizational	5	
PL-04(01)	Rules of Behavior Social Media and External Site/application Usage	include in the rules of behavior, restrictions on: a. Use of social media, social networking sites, and external sites/applications; b. Posting organizational information on public websites; and c. Use of organization-provided identifiers	Functional	Equal	Social Media & Social Networking Restrictions	HRS-05.2	Mechanisms exist to define rules of behavior that contain explicit restrictions on the use of social media and networking sites, posting	10	
PL-08	Security and Privacy Architectures	a. Develop security and privacy architectures for the system that: 1. Describe the requirements and approach to be taken for protecting the confidentiality, integrity, and availability of organizational information; 2. Describe the requirements and	Functional	Intersects With	Alignment With Enterprise Architecture	SEA-02	Mechanisms exist to develop an enterprise architecture, aligned with industry-recognized leading practices, with consideration for	5	
PL-10	Baseline Selection	Select a control baseline for the system.	Functional	Equal	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for	10	
PL-11	Baseline Tailoring	Tailor the selected control baseline by applying specified tailoring actions.	Functional	Equal	Baseline Tailoring	CFG-02.9	Mechanisms exist to allow baseline controls to be specialized or customized by applying a defined set of tailoring actions that are specific	10	
PM-01	Information Security Program Plan	a. Develop and disseminate an organization-wide information security program plan that: 1. Provides an overview of the requirements for the security program and a description of the security program management controls and common controls	Functional	Subset Of	Security, Compliance & Resilience Program (SCRPP)	GOV-01	Mechanisms exist to facilitate the implementation of security, compliance and resilience governance controls	10	

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Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Moderate O**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-mod>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
PM-01	Information Security Program Plan	a. Develop and disseminate an organization-wide information security program plan that:1. Provides an overview of the requirements for the security program and a description of the security program management controls and common controls	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRCP), including policies, standards and procedures	5	
PM-01	Information Security Program Plan	a. Develop and disseminate an organization-wide information security program plan that:1. Provides an overview of the requirements for the security program and a description of the security program management controls and common controls	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
PM-02	Information Security Program Leadership Role	Appoint a senior agency information security officer with the mission and resources to coordinate, develop, implement, and maintain an organization-wide information security program.	Functional	Intersects With	Assigned Security, Compliance & Resilience Responsibilities	GOV-04	Mechanisms exist to assign one or more qualified individuals with the mission and resources to centrally-manage, coordinate, develop	5	
PM-03	Information Security and Privacy Resources	a. Include the resources needed to implement the information security and privacy programs in capital planning and investment requests and document all exceptions to this requirement;b. Prepare documentation required for addressing	Functional	Equal	Security, Compliance & Resilience Resource Management	PRM-02	Mechanisms exist to address all capital planning and investment requests, including the resources needed to implement the Security	10	
PM-04	Plan of Action and Milestones Process	a. Implement a process to ensure that plans of action and milestones for the information security, privacy, and supply chain risk management programs and associated organizational systems:1. Are developed and maintained;2.	Functional	Intersects With	Vulnerability Remediation Process	VPM-02	Mechanisms exist to ensure that vulnerabilities are properly identified, tracked and remediated.	5	
PM-04	Plan of Action and Milestones Process	a. Implement a process to ensure that plans of action and milestones for the information security, privacy, and supply chain risk management programs and associated organizational systems:1. Are developed and maintained;2.	Functional	Intersects With	Capabilities Deficiency Tracking	IAO-05	Mechanisms exist to govern identified deficiencies (e.g., Plan of Action and Milestones (POA&M) or similar methodology) that formally	5	
PM-05	System Inventory	Develop and update [Assignment: organization-defined frequency] an inventory of organizational systems.	Functional	Intersects With	Asset Governance	AST-01	Mechanisms exist to facilitate an IT Asset Management (ITAM) program to implement and manage asset management controls.	5	
PM-05	System Inventory	Develop and update [Assignment: organization-defined frequency] an inventory of organizational systems.	Functional	Intersects With	Asset Inventories	AST-02	Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data Assets	5	
PM-06	Measures of Performance	Develop, monitor, and report on the results of information security and privacy measures of performance.	Functional	Intersects With	Assigned Security, Compliance & Resilience Responsibilities	GOV-04	Mechanisms exist to assign one or more qualified individuals with the mission and resources to centrally-manage, coordinate, develop	5	
PM-06	Measures of Performance	Develop, monitor, and report on the results of information security and privacy measures of performance.	Functional	Intersects With	Measures of Performance	GOV-05	Mechanisms exist to develop, report and monitor Security, Compliance & Resilience Program (SCRCP) measures of performance.	5	
PM-07	Enterprise Architecture	Develop and maintain an enterprise architecture with consideration for information security, privacy, and the resulting risk to organizational operations and assets, individuals, other organizations, and the Nation.	Functional	Intersects With	Alignment With Enterprise Architecture	SEA-02	Mechanisms exist to develop an enterprise architecture, aligned with industry-recognized leading practices, with consideration for	5	
PM-08	Critical Infrastructure Plan	Address information security and privacy issues in the development, documentation, and updating of a critical infrastructure and key resources protection plan.	Functional	Intersects With	Business Continuity Management System (BCMS)	BCD-01	Mechanisms exist to facilitate the implementation of contingency planning controls to help ensure	5	
PM-08	Critical Infrastructure Plan	Address information security and privacy issues in the development, documentation, and updating of a critical infrastructure and key resources protection plan.	Functional	Intersects With	Statutory, Regulatory & Contractual Compliance	CPL-01	Mechanisms exist to facilitate the identification and implementation of relevant statutory, regulatory and contractual controls.	5	
PM-09	Risk Management Strategy	a. Develops a comprehensive strategy to manage:1. Security risk to organizational operations and assets, individuals, other organizations, and the Nation associated with the operation and use of organizational systems; and2. Privacy risk to	Functional	Equal	Risk Management Program	RSK-01	Mechanisms exist to facilitate the implementation of strategic, operational and tactical risk management controls that are	10	
PM-10	Authorization Process	a. Manage the security and privacy state of organizational systems and the environments in which those systems operate through authorization processes;b. Designate individuals to fulfill specific roles and responsibilities within the	Functional	Equal	Information Assurance (IA) Operations	IAO-01	Mechanisms exist to facilitate the implementation of security, compliance and resilience assessment and authorization	10	
PM-11	Mission and Business Process Definition	a. Define organizational mission and business processes with consideration for information security and privacy and the resulting risk to organizational operations, organizational assets, individuals, other organizations, and the Nation; andb.	Functional	Equal	Business Process Definition	PRM-06	Mechanisms exist to define business processes with consideration for security, compliance and resilience that determines	10	
PM-12	Insider Threat Program	Implement an insider threat program that includes a cross-discipline insider threat incident handling team.	Functional	Equal	Insider Threat Program	THR-04	Mechanisms exist to implement an insider threat program that includes a cross-discipline insider threat incident handling team.	10	
PM-13	Security and Privacy Workforce	Establish a security and privacy workforce development and improvement program.	Functional	Intersects With	Defined Roles & Responsibilities	HRS-03	Mechanisms exist to define cybersecurity roles & responsibilities for all personnel.	5	
PM-13	Security and Privacy Workforce	Establish a security and privacy workforce development and improvement program.	Functional	Intersects With	Security, Compliance & Resilience-Minded Workforce	SAT-01	Mechanisms exist to facilitate the implementation of security workforce development and awareness	5	
PM-14	Testing, Training, and Monitoring	a. Implement a process for ensuring that organizational plans for conducting security and privacy testing, training, and monitoring activities associated with organizational systems:1. Are developed and maintained; and2. Continue to be executed;	Functional	Intersects With	Personal Data (PD) Control Testing, Training & Monitoring	PRI-08	Mechanisms exist to conduct testing, training and monitoring activities for Personal Data (PD) controls.	5	
PM-14	Testing, Training, and Monitoring	a. Implement a process for ensuring that organizational plans for conducting security and privacy testing, training, and monitoring activities associated with organizational systems:1. Are developed and maintained; and2. Continue to be executed;	Functional	Intersects With	Security, Compliance & Resilience Controls Oversight	CPL-02	Mechanisms exist to provide a security, compliance and resilience controls oversight function that reports to the organization's executive	5	
PM-15	Security and Privacy Groups and Associations	Establish and institutionalize contact with selected groups and associations within the security and privacy communities:a. To facilitate ongoing security and privacy education and training for organizational personnel;b. To maintain currency with	Functional	Intersects With	Threat Intelligence Program	THR-01	Mechanisms exist to implement a threat intelligence program that includes a cross-organization information-sharing capability that	5	
PM-15	Security and Privacy Groups and Associations	Establish and institutionalize contact with selected groups and associations within the security and privacy communities:a. To facilitate ongoing security and privacy education and training for organizational personnel;b. To maintain currency with	Functional	Intersects With	Contacts With Groups & Associations	GOV-07	Mechanisms exist to establish contact with selected groups and associations within the security, compliance and resilience	5	
PM-16	Threat Awareness Program	Implement a threat awareness program that includes a cross-organization information-sharing capability for threat intelligence.	Functional	Intersects With	Threat Intelligence Program	THR-01	Mechanisms exist to implement a threat intelligence program that includes a cross-organization information-sharing capability that	5	
PM-17	Protecting Controlled Unclassified Information on External Systems	a. Establish policy and procedures to ensure that requirements for the protection of controlled unclassified information that is processed, stored or transmitted on external systems, are implemented in accordance with applicable laws, executive	Functional	Equal	Protecting Sensitive / Regulated Data on External Technology Assets, Applications	DCH-13.3	Mechanisms exist to ensure that the requirements for the protection of sensitive and/or regulated data processed, stored, transmitted on	10	

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
PM-18	Privacy Program Plan	a. Develop and disseminate an organization-wide privacy program plan that provides an overview of the agency's privacy program, and:1. Includes a description of the structure of the privacy program and the resources dedicated to the privacy program.	Functional	Equal	Data Privacy Program	PRI-01	Mechanisms exist to facilitate the implementation and operation of data protection controls throughout the organization.	10	
PM-19	Privacy Program Leadership Role	Appoint a senior agency official for privacy with the authority, mission, accountability, and resources to coordinate, develop, and implement, applicable privacy requirements and manage privacy risks through the organization-wide privacy program.	Functional	Equal	Chief Privacy Officer (CPO)	PRI-01.1	Mechanisms exist to appoints a Chief Privacy Officer (CPO) or similar role, with the authority, mission, accountability and resources to	10	
PM-20	Dissemination of Privacy Program Information	Maintain a central resource webpage on the organization's principal public website that serves as a central source of information about the organization's privacy program and that: a. Ensures that the public has access to information about the organization's privacy policies on all external-facing websites, mobile applications, and other digital services, that: a. Are written in plain language and organized in a way that is easy to understand and navigate; b. Provide information	Functional	Equal	Dissemination of Data Privacy Program Information	PRI-01.3	Mechanisms exist to: (1) Ensure that the public has access to information about organizational data privacy activities and can	10	
PM-20(01)	Dissemination of Privacy Program Information Privacy Policies on Websites	Develop and post privacy policies on all external-facing websites, mobile applications, and other digital services, that: a. Are written in plain language and organized in a way that is easy to understand and navigate; b. Provide information	Functional	Equal	Data Privacy Notice	PRI-02	Mechanisms exist to: (1) Make data privacy notice(s) available to individuals upon first interaction with an organization and	10	
PM-21	Accounting of Disclosures	a. Develop and maintain an accurate accounting of disclosures of personally identifiable information, including:1. Date, nature, and purpose of each disclosure; and2. Name and address, or other contact information of the individual or	Functional	Equal	Accounting of Disclosures	PRI-14.1	Mechanisms exist to provide data subjects with an accounting of disclosures of their Personal Data (PD) controlled by	10	
PM-22	Personally Identifiable Information Quality Management	Develop and document organization-wide policies and procedures for: a. Reviewing for the accuracy, relevance, timeliness, and completeness of personally identifiable information across the information life cycle; b. Correcting or	Functional	Intersects With	Data Quality Management	PRI-10	Mechanisms exist to manage the quality, utility, objectivity, integrity and impact determination and de-identification of sensitive and/or	5	
PM-22	Personally Identifiable Information Quality Management	Develop and document organization-wide policies and procedures for: a. Reviewing for the accuracy, relevance, timeliness, and completeness of personally identifiable information across the information life cycle; b. Correcting or	Functional	Intersects With	Data Quality Operations	DCH-22	Mechanisms exist to check for Redundant, Obsolete/Outdated, Toxic or Trivial (ROT) data to ensure the accuracy, relevance, timeliness	5	
PM-23	Data Governance Body	Establish a Data Governance Body consisting of [Assignment: organization-defined roles] with [Assignment: organization-defined responsibilities].	Functional	Intersects With	Data Management Board	PRI-13	Mechanisms exist to establish a written charter for a Data Management Board (DMB) and assigned organization-defined roles	5	
PM-23	Data Governance Body	Establish a Data Governance Body consisting of [Assignment: organization-defined roles] with [Assignment: organization-defined responsibilities].	Functional	Intersects With	Data Quality Management	PRI-10	Mechanisms exist to manage the quality, utility, objectivity, integrity and impact determination and de-identification of sensitive and/or	5	
PM-23	Data Governance Body	Establish a Data Governance Body consisting of [Assignment: organization-defined roles] with [Assignment: organization-defined responsibilities].	Functional	Intersects With	Data Governance	GOV-10	Mechanisms exist to facilitate data governance to oversee the organization's policies, standards and procedures so that sensitive and/or	5	
PM-24	Data Integrity Board	Establish a Data Integrity Board to: a. Review proposals to conduct or participate in a matching program; and b. Conduct an annual review of all matching programs in which the agency has participated.	Functional	Intersects With	Data Governance	GOV-10	Mechanisms exist to facilitate data governance to oversee the organization's policies, standards and procedures so that sensitive and/or	5	
PM-24	Data Integrity Board	Establish a Data Integrity Board to: a. Review proposals to conduct or participate in a matching program; and b. Conduct an annual review of all matching programs in which the agency has participated.	Functional	Intersects With	Data Management Board	PRI-13	Mechanisms exist to establish a written charter for a Data Management Board (DMB) and assigned organization-defined roles	5	
PM-24	Data Integrity Board	Establish a Data Integrity Board to: a. Review proposals to conduct or participate in a matching program; and b. Conduct an annual review of all matching programs in which the agency has participated.	Functional	Intersects With	Data Quality Management	PRI-10	Mechanisms exist to manage the quality, utility, objectivity, integrity and impact determination and de-identification of sensitive and/or	5	
PM-24	Data Integrity Board	Establish a Data Integrity Board to: a. Review proposals to conduct or participate in a matching program; and b. Conduct an annual review of all matching programs in which the agency has participated.	Functional	Intersects With	Personal Data (PD) Accuracy & Integrity	PRI-05.2	Mechanisms exist to ensure the accuracy and relevance of Personal Data (PD) throughout the information lifecycle.	5	
PM-24	Data Integrity Board	Establish a Data Integrity Board to: a. Review proposals to conduct or participate in a matching program; and b. Conduct an annual review of all matching programs in which the agency has participated.	Functional	Intersects With	Computer Matching Agreements (CMA)	PRI-02.3	Mechanisms exist to publish Computer Matching Agreements (CMA) on the organization's public	5	
PM-24	Data Integrity Board	Establish a Data Integrity Board to: a. Review proposals to conduct or participate in a matching program; and b. Conduct an annual review of all matching programs in which the agency has participated.	Functional	Intersects With	Automated Data Management Processes	PRI-02.2	Automated mechanisms exist to adjust data that is able to be collected, received, processed, stored, transmitted, shared, updated	5	
PM-25	Minimization of Personally Identifiable Information Used in Testing, Training, and	a. Develop, document, and implement policies and procedures that address the use of personally identifiable information for internal testing, training, and research; b. Limit or minimize the amount of personally identifiable information used for internal	Functional	Intersects With	Usage Restrictions of Personal Data (PD)	PRI-05.4	Mechanisms exist to restrict collecting, receiving, processing, storing, transmitting, sharing and/or updating Personal Data (PD) to	5	
PM-25	Minimization of Personally Identifiable Information Used in Testing, Training, and	a. Develop, document, and implement policies and procedures that address the use of personally identifiable information for internal testing, training, and research; b. Limit or minimize the amount of personally identifiable information used for internal	Functional	Intersects With	Collection Minimization	END-13.3	Mechanisms exist to utilize sensors that are configured to minimize the collection of information about individuals.	5	
PM-25	Minimization of Personally Identifiable Information Used in Testing, Training, and	a. Develop, document, and implement policies and procedures that address the use of personally identifiable information for internal testing, training, and research; b. Limit or minimize the amount of personally identifiable information used for internal	Functional	Intersects With	Minimize Visitor Personal Data (PD)	PES-06.5	Mechanisms exist to minimize the collection of Personal Data (PD) contained in visitor access records.	5	
PM-25	Minimization of Personally Identifiable Information Used in Testing, Training, and	a. Develop, document, and implement policies and procedures that address the use of personally identifiable information for internal testing, training, and research; b. Limit or minimize the amount of personally identifiable information used for internal	Functional	Intersects With	Internal Use of Personal Data (PD) For Testing, Training and Research	PRI-05.1	Mechanisms exist to address the use of Personal Data (PD) for internal testing, training and research that:	5	
PM-25	Minimization of Personally Identifiable Information Used in Testing, Training, and	a. Develop, document, and implement policies and procedures that address the use of personally identifiable information for internal testing, training, and research; b. Limit or minimize the amount of personally identifiable information used for internal	Functional	Intersects With	Regulated Data In Testing, Training & Research	DCH-18.2	Mechanisms exist to minimize the use of sensitive and/or regulated data for research, testing, or training, in accordance with authorized	5	
PM-26	Complaint Management	Implement a process for receiving and responding to complaints, concerns, or questions from individuals about the organizational security and privacy practices that includes: a. Mechanisms that are easy to use and readily accessible by the	Functional	Intersects With	Data Subject Feedback Management	PRI-06.4	Mechanisms exist to maintain a process to efficiently and effectively respond to requests, complaints, concerns and suggestions from	5	
PM-26	Complaint Management	Implement a process for receiving and responding to complaints, concerns, or questions from individuals about the organizational security and privacy practices that includes: a. Mechanisms that are easy to use and readily accessible by the	Functional	Intersects With	Appeal Adverse Decision	PRI-06.3	Mechanisms exist to maintain a process for data subjects to appeal an adverse decision.	5	
PM-27	Privacy Reporting	a. Develop [Assignment: organization-defined privacy reports] and disseminate to: 1. [Assignment: organization-defined oversight bodies] to demonstrate accountability with statutory, regulatory and policy privacy mandates; and 2. [Assignment: a. Identify and document: 1. Assumptions affecting risk	Functional	Equal	Documenting Data Processing Activities	PRI-14	Mechanisms exist to document Personal Data (PD) processing activities that covers collection,	10	
PM-28	Risk Framing	a. Identify and document: 1. Assumptions affecting risk assessments, risk responses, and risk monitoring; 2. Constraints affecting risk assessments, risk responses, and risk monitoring; 3. Priorities and trade-offs considered by the	Functional	Equal	Risk Framing	RSK-01.1	Mechanisms exist to identify: (1) Assumptions affecting risk assessments, risk response and risk monitoring;	10	

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
PM-29	Risk Management Program Leadership Roles	a. Appoint a Senior Accountable Official for Risk Management to align organizational information security and privacy management processes with strategic, operational, and budgetary planning processes; and b. Establish a Risk Executive	Functional	Intersects With	Supply Chain Risk Management (SCRM) Plan	RSK-09	Mechanisms exist to develop a plan for Supply Chain Risk Management (SCRM) associated with the development activities	5	
PM-29	Risk Management Program Leadership Roles	a. Appoint a Senior Accountable Official for Risk Management to align organizational information security and privacy management processes with strategic, operational, and budgetary planning processes; and b. Establish a Risk Executive	Functional	Intersects With	Assigned Security, Compliance & Resilience Responsibilities	GOV-04	Mechanisms exist to assign one or more qualified individuals with the mission and resources to centrally manage, coordinate, develop	5	
PM-29	Risk Management Program Leadership Roles	a. Appoint a Senior Accountable Official for Risk Management to align organizational information security and privacy management processes with strategic, operational, and budgetary planning processes; and b. Establish a Risk Executive	Functional	Intersects With	Risk Management Program	RSK-01	Mechanisms exist to facilitate the implementation of strategic, operational and tactical risk management controls that are	5	
PM-30	Supply Chain Risk Management Strategy	a. Develop an organization-wide strategy for managing supply chain risks associated with the development, acquisition, maintenance, and disposal of systems, system components, and system services; b. Implement the supply chain risk	Functional	Equal	Supply Chain Risk Management (SCRM) Plan	RSK-09	Mechanisms exist to develop a plan for Supply Chain Risk Management (SCRM) associated with the development activities	10	
PM-30(01)	Supply Chain Risk Management Strategy Suppliers of Critical or Mission-essential	Identify, prioritize, and assess suppliers of critical or mission-essential technologies, products, and services.	Functional	Intersects With	Customized Development of Critical Components	TDA-12	Mechanisms exist to custom-develop critical system components, when Commercial Off The Shelf (COTS) solutions are unavailable	5	
PM-30(01)	Supply Chain Risk Management Strategy Suppliers of Critical or Mission-essential	Identify, prioritize, and assess suppliers of critical or mission-essential technologies, products, and services.	Functional	Intersects With	Criticality Analysis During Development	TDA-06.1	Mechanisms exist to require the developer of the Technology Asset, Application and/or Service (TAAS) to perform activities that are	5	
PM-30(01)	Supply Chain Risk Management Strategy Suppliers of Critical or Mission-essential	Identify, prioritize, and assess suppliers of critical or mission-essential technologies, products, and services.	Functional	Intersects With	Third-Party Criticality Assessments	TPM-02	Mechanisms exist to identify, prioritize and assess suppliers and partners of critical Technology Assets, Applications and/or Services	5	
PM-31	Continuous Monitoring Strategy	Develop an organization-wide continuous monitoring strategy and implement continuous monitoring programs that include: a. Establishing the following organization-wide metrics to be monitored: [Assignment: organization-defined metrics]; b. Analyze [Assignment: organization-defined systems or systems components] supporting mission essential services or functions to ensure that the information resources are being used consistent with their intended purpose	Functional	Subset Of	Continuous Monitoring	MON-01	Mechanisms exist to facilitate the implementation of enterprise-wide monitoring controls.	10	
PM-32	Purposing	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] personnel security policy that: a. Addresses	Functional	Equal	Purpose Validation	GOV-11	Mechanisms exist to monitor mission/business-critical Technology Assets, Applications and/or Services (TAAS) to ensure these resources are	10	
PS-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] personnel security policy that: a. Addresses	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
PS-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] personnel security policy that: a. Addresses	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures, at	5	
PS-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] personnel security policy that: a. Addresses	Functional	Subset Of	Human Resources Security Management	HRS-01	Mechanisms exist to facilitate the implementation of personnel security controls.	10	
PS-02	Position Risk Designation	a. Assign a risk designation to all organizational positions; b. Establish screening criteria for individuals filling those positions; and c. Review and update position risk designations [Assignment: organization-defined frequency]	Functional	Intersects With	Competency Requirements for Security-Related Positions	HRS-03.2	Mechanisms exist to ensure that all security-related positions are staffed by qualified individuals who have the necessary skill set	5	
PS-02	Position Risk Designation	a. Assign a risk designation to all organizational positions; b. Establish screening criteria for individuals filling those positions; and c. Review and update position risk designations [Assignment: organization-defined frequency]	Functional	Intersects With	Position Categorization	HRS-02	Mechanisms exist to manage personnel security risk by assigning a risk designation to all positions and establishing screening criteria for	5	
PS-03	Personnel Screening	a. Screen individuals prior to authorizing access to the system; and b. Rescreen individuals in accordance with [Assignment: organization-defined conditions requiring rescreening and, where rescreening is so indicated, the frequency of	Functional	Equal	Personnel Screening	HRS-04	Mechanisms exist to manage personnel security risk by screening individuals prior to authorizing access	10	
PS-04	Personnel Termination	Upon termination of individual employment: a. Disable system access within [Assignment: organization-defined time period]; b. Terminate or revoke any authenticators and credentials associated with the individual; c. Conduct exit	Functional	Equal	Personnel Termination	HRS-09	Mechanisms exist to govern the termination of individual employment.	10	
PS-05	Personnel Transfer	a. Review and confirm ongoing operational need for current logical and physical access authorizations to systems and facilities when individuals are reauthorized or transferred to other positions within the organization; b. Initiate [Assignment:	Functional	Equal	Personnel Transfer	HRS-08	Mechanisms exist to adjust logical and physical access authorizations to Technology Assets, Applications and/or Services (TAAS) and facilities	10	
PS-06	Access Agreements	a. Develop and document access agreements for organizational systems; b. Review and update the access agreements [Assignment: organization-defined frequency]; and c. Verify that individuals requiring access to organizational	Functional	Intersects With	Confidentiality Agreements	HRS-06.1	Mechanisms exist to require Non-Disclosure Agreements (NDAs) or similar confidentiality agreements that reflect the needs to protect data	5	
PS-06	Access Agreements	a. Develop and document access agreements for organizational systems; b. Review and update the access agreements [Assignment: organization-defined frequency]; and c. Verify that individuals requiring access to organizational	Functional	Intersects With	Access Agreements	HRS-06	Mechanisms exist to require internal and third-party users to sign appropriate access agreements prior to being granted access	5	
PS-07	External Personnel Security	a. Establish personnel security requirements, including security roles and responsibilities for external providers; b. Require external providers to comply with personnel security policies and procedures established by the organization; c.	Functional	Equal	Third-Party Personnel	HRS-10	Mechanisms exist to govern third-party personnel by reviewing and monitoring third-party security, compliance and/or resilience roles	10	
PS-08	Personnel Sanctions	a. Employ a formal sanctions process for individuals failing to comply with established information security and privacy policies and procedures; and b. Notify [Assignment: organization-defined personnel or roles] within [Assignment:	Functional	Equal	Personnel Sanctions	HRS-07	Mechanisms exist to sanction personnel failing to comply with established security policies, standards and procedures	10	
PS-09	Position Descriptions	Incorporate security and privacy roles and responsibilities into organizational position descriptions.	Functional	Equal	Defined Roles & Responsibilities	HRS-03	Mechanisms exist to define cybersecurity roles & responsibilities for all personnel.	10	
RA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] risk assessment policy that: a. Addresses	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures, at	5	
RA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] risk assessment policy that: a. Addresses	Functional	Subset Of	Risk Management Program	RSK-01	Mechanisms exist to facilitate the implementation of strategic, operational and tactical risk management controls that are	10	
RA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] risk assessment policy that: a. Addresses	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Moderate O**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-mod>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
RA-02	Security Categorization	a. Categorize the system and information it processes, stores, and transmits;b. Document the security categorization results, including supporting rationale, in the security plan for the system; and;c. Verify that the authorizing official or authorizing	Functional	Equal	Risk-Based Security Categorization	RSK-02	Mechanisms exist to categorize Technology Assets, Applications, Services and/or Data (TAASD) in accordance with applicable laws	10	
RA-03	Risk Assessment	a. Conduct a risk assessment, including:1. Identifying threats to and vulnerabilities in the system;2. Determining the likelihood and magnitude of harm from unauthorized access, use, disclosure, disruption, modification, or destruction of the	Functional	Intersects With	Functional Review Of Security, Compliance & Resilience Controls	CPL-03.2	Mechanisms exist to regularly review Technology Assets, Applications and/or Services (TAAS) for adherence to the organization's security	5	
RA-03	Risk Assessment	a. Conduct a risk assessment, including:1. Identifying threats to and vulnerabilities in the system;2. Determining the likelihood and magnitude of harm from unauthorized access, use, disclosure, disruption, modification, or destruction of the	Functional	Intersects With	Risk Assessment	RSK-04	Mechanisms exist to conduct recurring assessments of risk that includes the likelihood and	5	
RA-03(01)	Risk Assessment Supply Chain Risk Assessment	a. Assess supply chain risks associated with [Assignment: organization-defined systems, system components, and system services]; andb. Update the supply chain risk assessment [Assignment: organization-defined frequency]	Functional	Equal	Supply Chain Risk Assessment	RSK-09.1	Mechanisms exist to periodically assess supply chain risks associated with Technology Assets, Applications and/or Services (TAAS)	10	
RA-05	Vulnerability Monitoring and Scanning	a. Monitor and scan for vulnerabilities in the system and hosted applications [Assignment: organization-defined frequency and/or randomly in accordance with organization-defined process] and when new vulnerabilities potentially affecting the	Functional	Intersects With	Vulnerability Scanning	VPM-06	Mechanisms exist to detect vulnerabilities and configuration errors by routine vulnerability scanning of systems and	5	
RA-05	Vulnerability Monitoring and Scanning	a. Monitor and scan for vulnerabilities in the system and hosted applications [Assignment: organization-defined frequency and/or randomly in accordance with organization-defined process] and when new vulnerabilities potentially affecting the	Functional	Intersects With	Update Tool Capability	VPM-06.1	Mechanisms exist to update vulnerability scanning tools.	5	
RA-05(02)	Vulnerability Monitoring and Scanning Update Vulnerabilities to Be	Update the system vulnerabilities to be scanned [Selection (one or more); [Assignment: organization-defined frequency]; prior to a new scan; when new vulnerabilities are identified and reported]	Functional	Intersects With	Update Tool Capability	VPM-06.1	Mechanisms exist to update vulnerability scanning tools.	5	
RA-05(05)	Vulnerability Monitoring and Scanning Privileged Access	Implement privileged access authorization to [Assignment: organization-defined system components] for [Assignment: organization-defined vulnerability scanning activities].	Functional	Equal	Privileged Access	VPM-06.3	Mechanisms exist to implement privileged access authorization for selected vulnerability scanning activities	10	
RA-05(11)	Vulnerability Monitoring and Scanning Public Disclosure Program	Establish a public reporting channel for receiving reports of vulnerabilities in organizational systems and system components.	Functional	Equal	Vulnerability Disclosure Program (VDP)	THR-06	Mechanisms exist to establish a Vulnerability Disclosure Program (VDP) to assist with the secure development and maintenance of	10	
RA-07	Risk Response	Respond to findings from security and privacy assessments, monitoring, and audits in accordance with organizational risk tolerance.	Functional	Equal	Risk Response	RSK-06.1	Mechanisms exist to ensure proper risk response actions were performed to remediate findings from security, compliance and/or resilience-related	10	
RA-09	Criticality Analysis	Identify critical system components and functions by performing a criticality analysis for [Assignment: organization-defined systems, system components, or system services] at [Assignment: organization-defined decision points in the	Functional	Intersects With	Third-Party Criticality Assessments	TPM-02	Mechanisms exist to identify, prioritize and assess suppliers and partners of critical Technology	5	
SA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more); Organization-level; Mission/business process-level; System-level system and services acquisition policy that a	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
SA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more); Organization-level; Mission/business process-level; System-level system and services acquisition policy that a	Functional	Subset Of	Technology Development & Acquisition	TDA-01	Mechanisms exist to facilitate the implementation of tailored development and acquisition strategies, contract tools and	10	
SA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more); Organization-level; Mission/business process-level; System-level system and services acquisition policy that a	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Technology Assets, Applications and/or Services (TAAS) within the Secure Development Life Cycle	5	
SA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more); Organization-level; Mission/business process-level; System-level system and services acquisition policy that a	Functional	Intersects With	Secure Software Development Practices (SSDP)	TDA-06	Mechanisms exist to develop applications based on Secure Software Development Practices	5	
SA-02	Allocation of Resources	a. Determine the high-level information security and privacy requirements for the system or system service in mission and business process planning;b. Determine, document, and allocate the resources required to protect the system or a	Functional	Equal	Allocation of Resources	PRM-03	Mechanisms exist to identify and allocate resources for management, operational, technical and data protection requirements within	10	
SA-03	System Development Life Cycle	[Assignment: organization-defined system development life cycle] that incorporates information security and privacy considerations;b. Define and document information security	Functional	Intersects With	Technology Lifecycle Management	SEA-07.1	Mechanisms exist to manage the usable lifecycles of Technology Assets, Applications and/or Services (TAAS)	5	
SA-03	System Development Life Cycle	a. Acquire, develop, and manage the system using [Assignment: organization-defined system development life cycle] that incorporates information security and privacy considerations;b. Define and document information security	Functional	Intersects With	Secure Development Life Cycle (SDLC) Management	PRM-07	Mechanisms exist to ensure changes to Technology Assets, Applications and/or Services (TAAS) within the Secure Development Life Cycle	5	
SA-04	Acquisition Process	Include the following requirements, descriptions, and criteria, explicitly or by reference, using [Selection (one or more): standardized contract language; [Assignment: organization-defined contract language]] in the acquisition contract for the	Functional	Intersects With	Minimum Viable Product (MVP) Security Requirements	TDA-02	Mechanisms exist to design, develop and produce Technology Assets, Applications and/or Services (TAAS) in	5	
SA-04	Acquisition Process	Include the following requirements, descriptions, and criteria, explicitly or by reference, using [Selection (one or more): standardized contract language; [Assignment: organization-defined contract language]] in the acquisition contract for the	Functional	Intersects With	Third-Party Management	TPM-01	Mechanisms exist to facilitate the implementation of third-party management controls.	5	
SA-04	Acquisition Process	Include the following requirements, descriptions, and criteria, explicitly or by reference, using [Selection (one or more): standardized contract language; [Assignment: organization-defined contract language]] in the acquisition contract for the	Functional	Intersects With	Technology Development & Acquisition	TDA-01	Mechanisms exist to facilitate the implementation of tailored development and acquisition strategies, contract tools and	5	
SA-04	Acquisition Process	Include the following requirements, descriptions, and criteria, explicitly or by reference, using [Selection (one or more): standardized contract language; [Assignment: organization-defined contract language]] in the acquisition contract for the	Functional	Intersects With	Managing Changes To Third-Party Services	TPM-10	Mechanisms exist to control changes to services by suppliers, taking into account the criticality of business Technology Assets, Applications and/or Services (TAAS)	5	
SA-04(01)	Acquisition Process Functional Properties of Controls	Require the developer of the system, system component, or system service to provide a description of the functional properties of the controls to be implemented.	Functional	Intersects With	Functional Properties	TDA-04.1	Mechanisms exist to require software developers to provide information describing the functional properties of the security, compliance and	5	
SA-04(01)	Acquisition Process Functional Properties of Controls	Require the developer of the system, system component, or system service to provide a description of the functional properties of the controls to be implemented.	Functional	Intersects With	Network Diagrams & Data Flow Diagrams (DFDs)	AST-04	Mechanisms exist to maintain network architecture diagrams that: (1) Contain sufficient detail to assess the security of the network's	5	
SA-04(02)	Acquisition Process Design and Implementation Information for	Require the developer of the system, system component, or system service to provide design and implementation information for the controls that includes: [Selection (one or more): security-relevant external system interfaces; high-level	Functional	Intersects With	Network Diagrams & Data Flow Diagrams (DFDs)	AST-04	Mechanisms exist to maintain network architecture diagrams that: (1) Contain sufficient detail to assess the security of the network's	5	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Moderate O**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-mod>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
SA-04(02)	Acquisition Process Design and Implementation Information for	Require the developer of the system, system component, or system service to provide design and implementation information for the controls that includes: [Selection (one or more): security-relevant external system interfaces; high-level	Functional	Intersects With	Access to Program Source Code	TDA-20	Mechanisms exist to limit privileges to change software resident within software libraries.	5	
SA-04(02)	Acquisition Process Design and Implementation Information for	Require the developer of the system, system component, or system service to provide design and implementation information for the controls that includes: [Selection (one or more): security-relevant external system interfaces; high-level	Functional	Intersects With	Functional Properties	TDA-04.1	Mechanisms exist to require software developers to provide information describing the functional properties of the security compliance and	5	
SA-04(09)	Acquisition Process Functions, Ports, Protocols, and Services in Use	Require the developer of the system, system component, or system service to identify the functions, ports, protocols, and services intended for organizational use.	Functional	Equal	Ports, Protocols & Services in Use	TDA-02.1	Mechanisms exist to require the developers of Technology Assets, Applications and/or Services (TAAS)	10	
SA-04(10)	Acquisition Process Use of Approved PIV Products	Employ only information technology products on the FIPS 201-approved products list for Personal Identity Verification (PIV) capability implemented within organizational systems.	Functional	Intersects With	Information Assurance Enabled Products	TDA-02.2	Mechanisms exist to limit the use of commercially-provided Information Assurance (IA) and IA-enabled IT products to those products that have	5	
SA-04(12)	Acquisition Process Data Ownership	a. Include organizational data ownership requirements in the acquisition contract; andb. Require all data to be removed from the contractor's system and returned to the organization within [Assignment: organization-defined time frame].	Functional	Intersects With	Personal Data (PD) Lineage	PRI-09	Mechanisms exist to maintain a process to document the lineage of Personal Data (PD) by recording how the organization collects, receives,	5	
SA-04(12)	Acquisition Process Data Ownership	a. Include organizational data ownership requirements in the acquisition contract; andb. Require all data to be removed from the contractor's system and returned to the organization within [Assignment: organization-defined time frame].	Functional	Intersects With	Data Stewardship	DCH-01.1	Mechanisms exist to ensure data stewardship is assigned, documented and communicated.	5	
SA-04(12)	Acquisition Process Data Ownership	a. Include organizational data ownership requirements in the acquisition contract; andb. Require all data to be removed from the contractor's system and returned to the organization within [Assignment: organization-defined time frame].	Functional	Intersects With	Asset Ownership Assignment	AST-03	Mechanisms exist to ensure asset ownership responsibilities are assigned, tracked and managed at a team, individual, or responsible	5	
SA-05	System Documentation	a. Obtain or develop administrator documentation for the system, system component, or system service that describes:1. Secure configuration, installation, and operation of the system component or service?2. Effective use and	Functional	Intersects With	Documentation Requirements	TDA-04	Mechanisms exist to obtain, protect and distribute administrator documentation for Technology Assets, Applications and/or Services	5	
SA-05	System Documentation	a. Obtain or develop administrator documentation for the system, system component, or system service that describes:1. Secure configuration, installation, and operation of the system component or service?2. Effective use and	Functional	Intersects With	Asset Scope Classification	AST-04.1	Mechanisms exist to determine security, compliance and resilience control applicability by identifying, assigning and documenting the	5	
SA-08	Security and Privacy Engineering Principles	Apply the following systems security and privacy engineering principles in the specification, design, development, implementation, and modification of the system and system components:[Assignment: organization-defined systems	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications	5	
SA-08	Security and Privacy Engineering Principles	Apply the following systems security and privacy engineering principles in the specification, design, development, implementation, and modification of the system and system components:[Assignment: organization-defined systems	Functional	Intersects With	Secure Engineering Principles	SEA-01	Mechanisms exist to facilitate the implementation of industry-recognized security, compliance and	5	
SA-09	External System Services	a. Require that providers of external system services comply with organizational security and privacy requirements and employ the following controls:[Assignment: organization-defined controls]b. Define and document organizational	Functional	Equal	Third-Party Services	TPM-04	Mechanisms exist to mitigate the risks associated with third-party access to the organization's Technology Assets, Applications	10	
SA-09(02)	External System Services Identification of Functions, Ports	Require providers of the following external system services to identify the functions, ports, protocols, and other services required for the use of such services:[Assignment: organization-defined external system services]	Functional	Equal	External Connectivity Requirements - Identification of Ports, Protocols & Services	TPM-04.2	Mechanisms exist to require External Service Providers (ESPs) to identify and document the business need for	10	
SA-10	Developer Configuration Management	Require the developer of the system, system component, or system service to:a. Perform configuration management during system, component, or service [Selection (one or more): design; development; implementation; operation; disposal]b.	Functional	Equal	Developer Configuration Management	TDA-14	Mechanisms exist to require system developers and integrators to perform configuration management during system design, development	10	
SA-11	Developer Testing and Evaluation	Require the developer of the system, system component, or system service, at all post-design stages of the system development life cycle, to:a. Develop and implement a plan for ongoing security and privacy control assessments;b. Perform	Functional	Equal	Security, Compliance & Resilience Testing Throughout Development	TDA-09	Mechanisms exist to require system developers/integrators consult with security, compliance and/or resilience personnel to	10	
SA-15	Development Process, Standards, and Tools	a. Require the developer of the system, system component, or system service to follow a documented development process that:1. Explicitly addresses security and privacy requirements?2. Identifies the standards and tools used in the	Functional	Equal	Secure Software Development Practices (SSDP)	TDA-06	Mechanisms exist to develop applications based on Secure Software Development Practices (SSDP)	10	
SA-15(03)	Development Process, Standards, and Tools Criticality Analysis	Require the developer of the system, system component, or system service to perform a criticality analysis:a. At the following decision points in the system development life cycle:[Assignment: organization-defined decision points]in the	Functional	Equal	Criticality Analysis During Development	TDA-06.1	Mechanisms exist to require the developer of the Technology Asset, Application and/or Service (TAAS) to perform a criticality analysis at	10	
SA-22	Unsupported System Components	a. Replace system components when support for the components is no longer available from the developer, vendor, or manufacturer; orb. Provide the following options for alternative sources for continued support for unsupported	Functional	Intersects With	Unsupported Technology Assets, Applications and/or Services (TAAS)	TDA-17	Mechanisms exist to prevent unsupported Technology Assets, Applications and/or Services (TAAS)	5	
SA-22	Unsupported System Components	a. Replace system components when support for the components is no longer available from the developer, vendor, or manufacturer; orb. Provide the following options for alternative sources for continued support for unsupported	Functional	Intersects With	Alternate Sources for Continued Support	TDA-17.1	Mechanisms exist to provide in-house support or contract external providers for support with unsupported Technology Assets, Applications	5	
SC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level system and communications protection policy	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
SC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level system and communications protection policy	Functional	Subset Of	Network Security Controls (NSC)	NET-01	Mechanisms exist to develop, govern & update procedures to facilitate the implementation of Network Security Controls (NSC)	10	
SC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level system and communications protection policy	Functional	Subset Of	Secure Engineering Principles	SEA-01	Mechanisms exist to facilitate the implementation of industry-recognized security, compliance and	10	
SC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level system and communications protection policy	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPR), including policies, standards and procedures, at	5	
SC-02	Separation of System and User Functionality	Separate user functionality, including user interface services, from system management functionality.	Functional	Equal	Application Partitioning	SEA-03.2	Mechanisms exist to separate user functionality from system management functionality.	10	
SC-04	Information in Shared System Resources	Prevent unauthorized and unintended information transfer via shared system resources.	Functional	Equal	Information In Shared Resources	SEA-05	Mechanisms exist to prevent unauthorized and unintended information transfer via shared system resources	10	

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
SC-05	Denial-of-service Protection	a. [Selection (one): Protect against; Limit] the effects of the following types of denial-of-service events: [Assignment: organization-defined types of denial-of-service events]; andb. Employ the following controls to achieve the denial-of-service	Functional	Intersects With	Resource Priority	CAP-02	Mechanisms exist to control resource utilization of Technology Assets, Applications and/or Services (TAAS) that are susceptible to Denial of	5	
SC-05	Denial-of-service Protection	a. [Selection (one): Protect against; Limit] the effects of the following types of denial-of-service events: [Assignment: organization-defined types of denial-of-service events]; andb. Employ the following controls to achieve the denial-of-service	Functional	Intersects With	Capacity Planning	CAP-03	Mechanisms exist to conduct capacity planning so that necessary capacity for information processing, telecommunications and	5	
SC-05	Denial-of-service Protection	a. [Selection (one): Protect against; Limit] the effects of the following types of denial-of-service events: [Assignment: organization-defined types of denial-of-service events]; andb. Employ the following controls to achieve the denial-of-service	Functional	Intersects With	Capacity & Performance Management	CAP-01	Mechanisms exist to facilitate the implementation of capacity management controls to ensure optimal system performance to meet	5	
SC-05	Denial-of-service Protection	a. [Selection (one): Protect against; Limit] the effects of the following types of denial-of-service events: [Assignment: organization-defined types of denial-of-service events]; andb. Employ the following controls to achieve the denial-of-service	Functional	Intersects With	Denial of Service (DoS) Protection	NET-02.1	Automated mechanisms exist to protect against or limit the effects of denial of service attacks.	5	
SC-07	Boundary Protection	a. Monitor and control communications at the external managed interfaces to the system and at key internal managed interfaces within the system;b. Implement subnetworks for publicly accessible system components that are [Selection	Functional	Intersects With	Boundary Protection	NET-03	Mechanisms exist to monitor and control communications at the external network boundary and at key internal boundaries within the	5	
SC-07(03)	Boundary Protection Access Points	Limit the number of external network connections to the system.	Functional	Equal	Limit Network Connections	NET-03.1	Mechanisms exist to limit the number of concurrent external network connections to its Technology Assets, Applications and/or Services (TAAS)	10	
SC-07(04)	Boundary Protection External Telecommunications Services	a. Implement a managed interface for each external telecommunication service;b. Establish a traffic flow policy for each managed interface;c. Protect the confidentiality and integrity of the information being transmitted across each	Functional	Intersects With	External Telecommunications Services	NET-03.2	Mechanisms exist to maintain a managed interface for each external telecommunication service that protects the confidentiality and	5	
SC-07(05)	Boundary Protection Deny by Default — Allow by Exception	Deny network communications traffic by default and allow network communications traffic by exception [Selection (one or more): at managed interfaces; for [Assignment: organization-defined system]]	Functional	Intersects With	Deny Traffic by Default & Allow Traffic by Exception	NET-04.1	Mechanisms exist to configure firewall and router configurations to deny network traffic by default and allow network traffic by exception	5	
SC-07(07)	Boundary Protection Split Tunneling for Remote Devices	Prevent split tunneling for remote devices connecting to organizational systems unless the split tunnel is securely provisioned using [Assignment: organization-defined safeguards]	Functional	Equal	Split Tunneling	CFG-03.4	Mechanisms exist to prevent split tunneling for remote devices unless the split tunnel is securely provisioned using organization-	10	
SC-07(08)	Boundary Protection Route Traffic to Authenticated Proxy Servers	Route [Assignment: organization-defined internal communications traffic] to [Assignment: organization-defined external networks] through authenticated proxy servers at managed interfaces.	Functional	Intersects With	Route Internal Traffic to Proxy Servers	NET-18.1	Mechanisms exist to route internal communications traffic to external networks through organization- approved proxy servers at managed	5	
SC-07(08)	Boundary Protection Route Traffic to Authenticated Proxy Servers	Route [Assignment: organization-defined internal communications traffic] to [Assignment: organization-defined external networks] through authenticated proxy servers at managed interfaces.	Functional	Intersects With	DNS & Content Filtering	NET-18	Mechanisms exist to force Internet-bound network traffic through a proxy device (e.g., Policy Enforcement Point (PEP)) for URL content filtering and	5	
SC-07(18)	Boundary Protection Fail Secure	Prevent systems from entering insecure states in the event of an operational failure of a boundary protection device.	Functional	Intersects With	Secure Engineering Principles	SEA-01	Mechanisms exist to facilitate the implementation of industry-recognized security, compliance and	5	
SC-07(28)	Boundary Protection Connections to Public Networks	Prohibit the direct connection of [Assignment: organization-defined system] to a public network.	Functional	Equal	Direct Internet Access Restrictions	NET-06.5	Mechanisms exist to prohibit, or strictly-control, Internet access from sensitive and/or regulated data enclaves (secure zones).	10	
SC-07(29)	Boundary Protection Separate Subnets to Isolate Functions	Implement [Selection (one): physically; logically] separate subnetworks to isolate the following critical system components and functions: [Assignment: organization-defined critical system components and functions]	Functional	Intersects With	Cloud Infrastructure Security Subnet	CLD-03	Mechanisms exist to host security-specific technologies in a dedicated subnet.	5	
SC-08	Transmission Confidentiality and Integrity	Protect the [Selection (one or more): confidentiality; integrity] of transmitted information.	Functional	Intersects With	Transmission Confidentiality	CRY-03	Cryptographic mechanisms exist to protect the confidentiality of data being transmitted.	5	
SC-08	Transmission Confidentiality and Integrity	Protect the [Selection (one or more): confidentiality; integrity] of transmitted information.	Functional	Intersects With	Transmission Integrity	CRY-04	Cryptographic mechanisms exist to protect the integrity of data being transmitted.	5	
SC-08(01)	Transmission Confidentiality and Integrity Cryptographic	Implement cryptographic mechanisms to [Selection (one or more): prevent unauthorized disclosure of information; detect changes to information] during transmission.	Functional	Intersects With	Alternate Physical Protection	CRY-01.1	Cryptographic mechanisms exist to prevent unauthorized disclosure of information as an alternative to physical safeguards.	5	
SC-08(01)	Transmission Confidentiality and Integrity Cryptographic	Implement cryptographic mechanisms to [Selection (one or more): prevent unauthorized disclosure of information; detect changes to information] during transmission.	Functional	Intersects With	Use of Cryptographic Controls	CRY-01	Mechanisms exist to facilitate the implementation of cryptographic protections controls using known public standards and trusted	5	
SC-08(01)	Transmission Confidentiality and Integrity Cryptographic	Implement cryptographic mechanisms to [Selection (one or more): prevent unauthorized disclosure of information; detect changes to information] during transmission.	Functional	Intersects With	Transmission Confidentiality	CRY-03	Cryptographic mechanisms exist to protect the confidentiality of data being transmitted.	5	
SC-12	Cryptographic Key Establishment and Management	Establish and manage cryptographic keys when cryptography is employed within the system in accordance with the following key management requirements: [Assignment: organization-defined requirements for key generation, distribution, storage,	Functional	Intersects With	Public Key Infrastructure (PKI)	CRY-08	Mechanisms exist to securely implement an internal Public Key Infrastructure (PKI) infrastructure or obtain PKI services from a reputable	5	
SC-13	Cryptographic Protection	a. Determine the [Assignment: organization-defined cryptographic uses]; andb. Implement the following types of cryptography required for each specified cryptographic use: [Assignment: organization-defined types of cryptography for	Functional	Intersects With	Encrypting Data At Rest	CRY-05	Cryptographic mechanisms exist to prevent unauthorized disclosure of data at rest.	5	
SC-13	Cryptographic Protection	a. Determine the [Assignment: organization-defined cryptographic uses]; andb. Implement the following types of cryptography required for each specified cryptographic use: [Assignment: organization-defined types of cryptography for	Functional	Intersects With	Export-Controlled Cryptography	CRY-01.2	Mechanisms exist to address the exporting of cryptographic technologies in compliance with relevant statutory and regulatory	5	
SC-13	Cryptographic Protection	a. Determine the [Assignment: organization-defined cryptographic uses]; andb. Implement the following types of cryptography required for each specified cryptographic use: [Assignment: organization-defined types of cryptography for	Functional	Intersects With	Use of Cryptographic Controls	CRY-01	Mechanisms exist to facilitate the implementation of cryptographic protections controls using known public standards and trusted	5	
SC-15	Collaborative Computing Devices and Applications	a. Prohibit remote activation of collaborative computing devices and applications with the following exceptions: [Assignment: organization-defined exceptions where remote activation is to be allowed]; andb. Provide an explicit indication	Functional	Intersects With	Collaborative Computing Devices	END-14	Mechanisms exist to unplug or prohibit the remote activation of collaborative computing devices with the following exceptions:	5	
SC-17	Public Key Infrastructure Certificates	a. Issue public key certificates under an [Assignment: organization-defined certificate policy] or obtain public key certificates from an approved service provider; andb. Include only approved trust anchors in trust stores or certificate stores	Functional	Intersects With	Public Key Infrastructure (PKI)	CRY-08	Mechanisms exist to securely implement an internal Public Key Infrastructure (PKI) infrastructure or obtain PKI services from a reputable	5	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Moderate O**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-mod>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
SC-18	Mobile Code	a. Define acceptable and unacceptable mobile code and mobile code technologies; and b. Authorize, monitor, and control the use of mobile code within the system.	Functional	Intersects With	Mobile Code	END-10	Mechanisms exist to address mobile code / operating system-independent applications.	5	
SC-20	Secure Name/address Resolution Service (authoritative Source)	a. Provide additional data origin authentication and integrity verification artifacts along with the authoritative name resolution data the system returns in response to external name/address resolution queries; and b. Provide the means to	Functional	Intersects With	Domain Name Service (DNS) Resolution	NET-10	Mechanisms exist to ensure Domain Name Service (DNS) resolution is designed, implemented and managed to protect the security of name /	5	
SC-21	Secure Name/address Resolution Service (recursive or Caching Resolver)	Request and perform data origin authentication and data integrity verification on the name/address resolution responses the system receives from authoritative sources.	Functional	Equal	Secure Name / Address Resolution Service (Recursive or Caching Resolver)	NET-10.2	Mechanisms exist to perform data origin authentication and data integrity verification on the Domain Name Service (DNS) resolution	10	
SC-22	Architecture and Provisioning for Name/address Resolution Service	Ensure the systems that collectively provide name/address resolution service for an organization are fault-tolerant and implement internal and external role separation.	Functional	Equal	Architecture & Provisioning for Name / Address Resolution Service	NET-10.1	Mechanisms exist to ensure systems that collectively provide Domain Name Service (DNS) resolution service are fault-tolerant and	10	
SC-23	Session Authenticity	Protect the authenticity of communications sessions.	Functional	Equal	Session Integrity	NET-09	Mechanisms exist to protect the authenticity and integrity of communications sessions.	10	
SC-24	Fail in Known State	Fail to a [Assignment: organization-defined known system state] for the following failures on the indicated components while preserving [Assignment: organization-defined system state information] in failure: [Assignment: list of organization-	Functional	Intersects With	Fail Secure	SEA-07.2	Mechanisms exist to enable systems to fail to an organization-defined known-state for types of failures, preserving system state information	5	
SC-28	Protection of Information at Rest	Protect the [Selection (one or more): confidentiality; integrity] of the following information at rest: [Assignment: organization-defined information at rest].	Functional	Intersects With	Endpoint Protection Measures	END-02	Mechanisms exist to protect the confidentiality, integrity, availability and safety of endpoint devices.	5	
SC-28	Protection of Information at Rest	Protect the [Selection (one or more): confidentiality; integrity] of the following information at rest: [Assignment: organization-defined information at rest].	Functional	Intersects With	Encrypting Data At Rest	CRY-05	Cryptographic mechanisms exist to prevent unauthorized disclosure of data at rest.	5	
SC-28(01)	Protection of Information at Rest Cryptographic Protection	Implement cryptographic mechanisms to prevent unauthorized disclosure and modification of the following information at rest on [Assignment: organization-defined system components or media]: [Assignment: organization-defined information]	Functional	Intersects With	Cryptographic Protection	BCD-11.4	Cryptographic mechanisms exist to prevent the unauthorized disclosure and/or modification of backup information	5	
SC-28(01)	Protection of Information at Rest Cryptographic Protection	Implement cryptographic mechanisms to prevent unauthorized disclosure and modification of the following information at rest on [Assignment: organization-defined system components or media]: [Assignment: organization-defined information]	Functional	Intersects With	Encrypting Data At Rest	CRY-05	Cryptographic mechanisms exist to prevent unauthorized disclosure of data at rest.	5	
SC-28(01)	Protection of Information at Rest Cryptographic Protection	Implement cryptographic mechanisms to prevent unauthorized disclosure and modification of the following information at rest on [Assignment: organization-defined system components or media]: [Assignment: organization-defined information]	Functional	Intersects With	Transmission Integrity	CRY-04	Cryptographic mechanisms exist to protect the integrity of data being transmitted.	5	
SC-28(01)	Protection of Information at Rest Cryptographic Protection	Implement cryptographic mechanisms to prevent unauthorized disclosure and modification of the following information at rest on [Assignment: organization-defined system components or media]: [Assignment: organization-defined information]	Functional	Intersects With	Encrypting Data In Storage Media	DCH-07.2	Cryptographic mechanisms exist to protect the confidentiality and integrity of information stored on digital media during transport outside	5	
SC-39	Process Isolation	Maintain a separate execution domain for each executing system process.	Functional	Equal	Process Isolation	SEA-04	Mechanisms exist to implement a separate execution domain for each executing process.	10	
SC-41	Port and I/O Device Access	[Selection (one): Physically; Logically] disable or remove [Assignment: organization-defined connection ports or input/output devices] on the following systems or system components: [Assignment: organization-defined systems or	Functional	Equal	Port & Input / Output (I/O) Device Access	END-12	Mechanisms exist to physically disable or remove unnecessary connection ports or input/output devices from sensitive systems	10	
SC-45	System Time Synchronization	Synchronize system clocks within and between systems and system components.	Functional	Intersects With	Synchronization With Authoritative Time Source	MON-07.1	Mechanisms exist to synchronize internal system clocks with an authoritative time source.	5	
SI-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level system and information integrity policy that a	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures at	5	
SI-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level system and information integrity policy that a	Functional	Subset Of	Secure Engineering Principles	SEA-01	Mechanisms exist to facilitate the implementation of industry-recognized security, compliance and resilience practices in the	10	
SI-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level system and information integrity policy that a	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
SI-02	Flaw Remediation	a. Identify, report, and correct system flaws; b. Test software and firmware updates related to flaw remediation for effectiveness and potential side effects before installation; c. Install security-relevant software and firmware updates within	Functional	Intersects With	Vulnerability & Patch Management Program (VPMP)	VPM-01	Mechanisms exist to facilitate the implementation and monitoring of vulnerability management controls.	5	
SI-02	Flaw Remediation	a. Identify, report, and correct system flaws; b. Test software and firmware updates related to flaw remediation for effectiveness and potential side effects before installation; c. Install security-relevant software and firmware updates within	Functional	Intersects With	Software & Firmware Patching	VPM-05	Mechanisms exist to conduct software patching for all deployed Technology Assets, Applications and/or Services (TAAS), including	5	
SI-02	Flaw Remediation	a. Identify, report, and correct system flaws; b. Test software and firmware updates related to flaw remediation for effectiveness and potential side effects before installation; c. Install security-relevant software and firmware updates within	Functional	Intersects With	Automatic Antimalware Signature Updates	END-04.1	Automated mechanisms exist to update antimalware technologies, including signature definitions.	5	
SI-02(02)	Flaw Remediation Automated Flaw Remediation Status	Determine if system components have applicable security-relevant software and firmware updates installed using [Assignment: organization-defined automated mechanisms] [Assignment: organization-defined frequency]	Functional	Intersects With	Automated Remediation Status	VPM-05.2	Automated mechanisms exist to determine the state of system components with regard to flaw remediation	5	
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code; b. Automatically update malicious code protection	Functional	Intersects With	Software & Firmware Patching	VPM-05	Mechanisms exist to conduct software patching for all deployed Technology Assets, Applications and/or Services (TAAS), including	5	
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code; b. Automatically update malicious code protection	Functional	Intersects With	Vulnerability & Patch Management Program (VPMP)	VPM-01	Mechanisms exist to facilitate the implementation and monitoring of vulnerability management controls.	5	
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code; b. Automatically update malicious code protection	Functional	Intersects With	Malicious Code Protection (Anti-Malware)	END-04	Mechanisms exist to utilize antimalware technologies to detect and eradicate malicious code.	5	

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
SI-03	Malicious Code Protection	a. Implement (Selection (one or more): signature based; non-signature based) malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code;b. Automatically update malicious code protection	Functional	Intersects With	Heuristic / Nonsignature-Based Detection	END-04.4	Mechanisms exist to utilize heuristic / nonsignature-based antimalware detection capabilities.	5	
SI-03	Malicious Code Protection	a. Implement (Selection (one or more): signature based; non-signature based) malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code;b. Automatically update malicious code protection	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5	
SI-03	Malicious Code Protection	a. Implement (Selection (one or more): signature based; non-signature based) malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code;b. Automatically update malicious code protection	Functional	Intersects With	Automatic Antimalware Signature Updates	END-04.1	Automated mechanisms exist to update antimalware technologies, including signature definitions.	5	
SI-03	Malicious Code Protection	a. Implement (Selection (one or more): signature based; non-signature based) malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code;b. Automatically update malicious code protection	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5	
SI-04	System Monitoring	a. Monitor the system to detect:1. Attacks and indicators of potential attacks in accordance with the following monitoring objectives: [Assignment: organization-defined monitoring objectives]; and2. Unauthorized local, network, and remote	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5	
SI-04	System Monitoring	a. Monitor the system to detect:1. Attacks and indicators of potential attacks in accordance with the following monitoring objectives: [Assignment: organization-defined monitoring objectives]; and2. Unauthorized local, network, and remote	Functional	Intersects With	Centralized Collection of Security Event Logs	MON-02	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support the centralized collection of security-	5	
SI-04	System Monitoring	a. Monitor the system to detect:1. Attacks and indicators of potential attacks in accordance with the following monitoring objectives: [Assignment: organization-defined monitoring objectives]; and2. Unauthorized local, network, and remote	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5	
SI-04	System Monitoring	a. Monitor the system to detect:1. Attacks and indicators of potential attacks in accordance with the following monitoring objectives: [Assignment: organization-defined monitoring objectives]; and2. Unauthorized local, network, and remote	Functional	Intersects With	Continuous Monitoring	MON-01	Mechanisms exist to facilitate the implementation of enterprise-wide monitoring controls.	5	
SI-04(02)	System Monitoring Automated Tools and Mechanisms for Real-time Analysis	Employ automated tools and mechanisms to support near real-time analysis of events.	Functional	Equal	Automated Tools for Real-Time Analysis	MON-01.2	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support near real-time analysis and incident	10	
SI-04(04)	System Monitoring Inbound and Outbound Communications	a. Determine criteria for unusual or unauthorized activities or conditions for inbound and outbound communications traffic;b. Monitor inbound and outbound communications traffic [Assignment: organization-defined frequency] for Alert [Assignment: organization-defined personnel or roles] when the following system-generated indications of compromise or potential compromise occur: [Assignment: organization-defined compromise indicators]	Functional	Equal	Inbound & Outbound Communications Traffic	MON-01.3	Mechanisms exist to continuously monitor inbound and outbound communications traffic for unusual or unauthorized activities or conditions.	10	
SI-04(05)	System Monitoring System-generated Alerts	a. Receive system security alerts, advisories, and directives from [Assignment: organization-defined external organizations] on an ongoing basis;b. Generate internal security alerts, advisories, and directives as deemed necessary;c.	Functional	Equal	System Generated Alerts	MON-01.4	Mechanisms exist to generate, monitor, correlate and respond to alerts from physical, cybersecurity, data protection, and availability	10	
SI-05	Security Alerts, Advisories, and Directives	a. Receive system security alerts, advisories, and directives from [Assignment: organization-defined external organizations] on an ongoing basis;b. Generate internal security alerts, advisories, and directives as deemed necessary;c.	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5	
SI-05	Security Alerts, Advisories, and Directives	a. Receive system security alerts, advisories, and directives from [Assignment: organization-defined external organizations] on an ongoing basis;b. Generate internal security alerts, advisories, and directives as deemed necessary;c.	Functional	Intersects With	Threat Intelligence Feeds	THR-03	Mechanisms exist to maintain situational awareness of vulnerabilities and evolving threats by leveraging the knowledge of attacker	5	
SI-05	Security Alerts, Advisories, and Directives	a. Receive system security alerts, advisories, and directives from [Assignment: organization-defined external organizations] on an ongoing basis;b. Generate internal security alerts, advisories, and directives as deemed necessary;c.	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5	
SI-07	Software, Firmware, and Information Integrity	a. Employ integrity verification tools to detect unauthorized changes to the following software, firmware, and information: [Assignment: organization-defined software, firmware, and information]; andb. Take the following actions when	Functional	Intersects With	Endpoint File Integrity Monitoring (FIM)	END-06	Mechanisms exist to utilize File Integrity Monitor (FIM), or similar technologies, to detect and report on unauthorized changes to selected	5	
SI-07	Software, Firmware, and Information Integrity	a. Employ integrity verification tools to detect unauthorized changes to the following software, firmware, and information: [Assignment: organization-defined software, firmware, and information]; andb. Take the following actions when	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5	
SI-07	Software, Firmware, and Information Integrity	a. Employ integrity verification tools to detect unauthorized changes to the following software, firmware, and information: [Assignment: organization-defined software, firmware, and information]; andb. Take the following actions when	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5	
SI-07(01)	Software, Firmware, and Information Integrity Integrity Checks	Perform an integrity check of [Assignment: organization-defined software, firmware, and information] [Selection (one or more): at startup; at [Assignment: organization-defined transitional states or security-relevant events]; [Assignment: incorporate the detection of the following unauthorized changes into the organizational incident response capability: [Assignment: organization-defined security-relevant changes to the system]	Functional	Equal	Integrity Checks	END-06.1	Mechanisms exist to validate configurations through integrity checking of software and firmware.	10	
SI-07(07)	Software, Firmware, and Information Integrity Integration of Detection and	a. Employ spam protection mechanisms at system entry and exit points to detect and act on unsolicited messages; andb. Update spam protection mechanisms when new releases are available in accordance with organizational configuration	Functional	Equal	Endpoint Detection & Response (EDR)	END-06.2	Mechanisms exist to detect and respond to unauthorized configuration changes as cybersecurity incidents.	10	
SI-08	Spam Protection	a. Employ spam protection mechanisms at system entry and exit points to detect and act on unsolicited messages; andb. Update spam protection mechanisms when new releases are available in accordance with organizational configuration	Functional	Equal	Phishing & Spam Protection	END-08	Mechanisms exist to utilize anti-phishing and spam protection technologies to detect and take action on unsolicited messages.	10	
SI-10	Information Input Validation	Check the validity of the following information inputs: [Assignment: organization-defined information inputs to the system].	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5	
SI-10	Information Input Validation	Check the validity of the following information inputs: [Assignment: organization-defined information inputs to the system].	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5	
SI-11	Error Handling	a. Generate error messages that provide information necessary for corrective actions without revealing information that could be exploited; andb. Reveal error messages only to [Assignment: organization-defined personnel or roles]	Functional	Equal	Error Handling	TDA-19	Mechanisms exist to handle error conditions by: (1) Identifying potentially security-relevant error conditions.	10	
SI-12	Information Management and Retention	Manage and retain information within the system and information output from the system in accordance with applicable laws, executive orders, directives, regulations, policies, standards, guidelines and operational requirements.	Functional	Intersects With	Media & Data Retention	DCH-18	Mechanisms exist to retain media and data in accordance with applicable statutory, regulatory and contractual obligations.	5	
SI-12	Information Management and Retention	Manage and retain information within the system and information output from the system in accordance with applicable laws, executive orders, directives, regulations, policies, standards, guidelines and operational requirements.	Functional	Intersects With	Personal Data (PD) Retention & Disposal	PRI-05	Mechanisms exist to: (1) Retain Personal Data (PD), including metadata, for an organization-defined time period to	5	

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Moderate O**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3-mod>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes
SI-16	Memory Protection	Implement the following controls to protect the system memory from unauthorized code execution: [Assignment: organization-defined controls].	Functional	Equal	Memory Protection	SEA-10	Mechanisms exist to implement security safeguards to protect system memory from unauthorized code execution.	10	
SI-17	Fail-safe Procedures	Implement the indicated fail-safe procedures when the indicated failures occur: [Assignment: organization-defined list of failure conditions and associated fail-safe procedures].	Functional	Equal	Fail Safe	SEA-07.3	Mechanisms exist to implement fail-safe procedures when failure conditions occur.	10	
SR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] supply chain risk management policy that: a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] supply chain risk management policy that: a.	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCR), including policies, standards and procedures necessary for secure, compliant and resilient	5	
SR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] supply chain risk management policy that: a.	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5	
SR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] supply chain risk management policy that: a.	Functional	Subset Of	Third-Party Management	TPM-01	Mechanisms exist to facilitate the implementation of third-party management controls.	10	
SR-02	Supply Chain Risk Management Plan	a. Develop a plan for managing supply chain risks associated with the research and development, design, manufacturing, acquisition, delivery, integration, operations and maintenance, and disposal of the following systems, system components or	Functional	Intersects With	Supply Chain Risk Management (SCRM) Plan	RSK-09	Mechanisms exist to develop a plan for Supply Chain Risk Management (SCRM) associated with the development, acquisition,	5	
SR-02	Supply Chain Risk Management Plan	a. Develop a plan for managing supply chain risks associated with the research and development, design, manufacturing, acquisition, delivery, integration, operations and maintenance, and disposal of the following systems, system components or	Functional	Intersects With	Supply Chain Risk Management (SCRM)	TPM-03	Mechanisms exist to: (1) Evaluate security risks and threats associated with Technology Assets, Acquisition and/or Services (TAAC)	5	
SR-02(01)	Supply Chain Risk Management Plan Establish SCRM Team	Establish a supply chain risk management team consisting of [Assignment: organization-defined personnel, roles, and responsibilities] to lead and support the following SCRM activities: [Assignment: organization-defined supply chain risk	Functional	Intersects With	Supply Chain Risk Management (SCRM)	TPM-03	Mechanisms exist to: (1) Evaluate security risks and threats associated with Technology Assets, Acquisition and/or Services (TAAC)	5	
SR-03	Supply Chain Controls and Processes	a. Establish a process or processes to identify and address weaknesses or deficiencies in the supply chain elements and processes of [Assignment: organization-defined system or system component] in coordination with [Assignment:	Functional	Equal	Processes To Address Weaknesses or Deficiencies	TPM-03.3	Mechanisms exist to address identified weaknesses or deficiencies in the security of the supply chain	10	
SR-05	Acquisition Strategies, Tools, and Methods	Employ the following acquisition strategies, contract tools, and procurement methods to protect against, identify, and mitigate supply chain risks: [Assignment: organization-defined acquisition strategies, contract tools, and procurement	Functional	Intersects With	Acquisition Strategies, Tools & Methods	TPM-03.1	Mechanisms exist to utilize tailored acquisition strategies, contract tools and procurement methods for the purchase of critical technology	5	
SR-05(01)	Acquisition Strategies, Tools, and Methods Adequate Supply	Employ the following controls to ensure an adequate supply of [Assignment: organization-defined critical system components]: [Assignment: organization-defined controls].	Functional	Equal	Adequate Supply	TPM-03.4	Mechanisms exist to develop and implement a spare parts strategy to ensure that an adequate supply of critical components is available to	10	
SR-06	Supplier Assessments and Reviews	Assess and review the supply chain-related risks associated with suppliers or contractors and the system, system component, or system service they provide [Assignment: organization-defined frequency]	Functional	Intersects With	Review of Third-Party Services	TPM-08	Mechanisms exist to monitor, regularly review and assess External Service Providers (ESPs) for compliance with established	5	
SR-08	Notification Agreements	Establish agreements and procedures with entities involved in the supply chain for the [Selection (one or more): notification of supply chain compromises; results of assessments or audits:	Functional	Equal	Security Compromise Notification Agreements	TPM-05.1	Mechanisms exist to compel External Service Providers (ESPs) to provide notification of actual or potential	10	
SR-10	Inspection of Systems or Components	Inspect the following systems or system components [Selection (one or more): at random; at [Assignment: organization-defined frequency], upon [Assignment: organization-defined indications of need for inspection]] to	Functional	Intersects With	Product Tampering and Counterfeiting (PTC)	TDA-11	Mechanisms exist to maintain awareness of component authenticity by developing and implementing Product Tampering and	5	
SR-10	Inspection of Systems or Components	Inspect the following systems or system components [Selection (one or more): at random; at [Assignment: organization-defined frequency], upon [Assignment: organization-defined indications of need for inspection]] to	Functional	Intersects With	Technology Asset Inspections	AST-15.1	Mechanisms exist to physically and logically inspect critical technology assets to detect evidence of	5	
SR-11	Component Authenticity	a. Develop and implement anti-counterfeit policy and procedures that include the means to detect and prevent counterfeit components from entering the system; andb. Report counterfeit system components to [Selection (one or	Functional	Intersects With	Product Tampering and Counterfeiting (PTC)	TDA-11	Mechanisms exist to maintain awareness of component authenticity by developing and implementing Product Tampering and	5	
SR-11(01)	Component Authenticity Anti-counterfeit Training	Train [Assignment: organization-defined personnel or roles] to detect counterfeit system components (including hardware, software, and firmware).	Functional	Equal	Anti-Counterfeit Training	TDA-11.1	Mechanisms exist to train personnel to detect counterfeit system components, including hardware, software and firmware.	10	
SR-11(02)	Component Authenticity Configuration Control for Component	Maintain configuration control over the following system components awaiting service or repair and serviced or repaired components awaiting return to service: [Assignment: organization-defined system components].	Functional	Equal	Maintain Configuration Control During Maintenance	MNT-07	Mechanisms exist to maintain proper physical security and configuration control over technology assets awaiting service or repair.	10	
SR-12	Component Disposal	Dispose of [Assignment: organization-defined data, documentation, tools, or system components] using the following techniques and methods: [Assignment: organization-defined techniques and methods].	Functional	Intersects With	Secure Disposal, Destruction or Re-Use of Equipment	AST-09	Mechanisms exist to securely dispose of, destroy or repurpose system components using organization-defined techniques and methods to	5	